

SESSION

4



Arithmetic II

Learning Objectives

1. Profit, Loss, Discount
2. SI - CI
3. Work, Time, Rate
3. Time, Speed and Distance

SESSION 04 – Arithmetic II

by PANKAJ BURHANPURKAR



SESSION 04 – Arithmetic II



REVISION

1. Find 33.33% of 900
2. What is 18.18% of 143
3. 25 is what % of 200
4. 80 is what % of 400
5. Find 31% of 910
6. Find the 30% increased value 440
7. Find the 25% reduced value of 760

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Cases						
	Case I and II (P and L)		Case III (MP)	Case IV (Cheating)	Case V (Malfunction)	Case VI (Adulteration)
Quantity	100 kg		100 kg	100 kg/ 900 kg	100 kg/ 1111.11 kg	100 kg/110 kg
Cost Price (CP)	Rs. 1000		Rs. 1000	Rs. 1000/ Rs. 900	Rs. 1000/ Rs. 1111.11	Rs. 1000/1100
Marked Price (MP)	–		Rs. 1300	–	–	–
Discount	–		10%	–	–	–
Selling Price (SP)	Rs. 1100	Rs. 900	Rs. 1170	Rs. 1000	Rs. 1000	Rs. 1000
Profit	Rs. 100	–	Rs. 170	Rs. 100	–	Rs. 100
Loss	–	Rs. 100	–	–	111.11	–
Profit %	10%	–	17%	11.11%	–	10%
Loss %	–	10%	–	–	9.99%	–
Margin %	9.09%	–	14.52%	10%	11.11%	9.09%
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Important Formulae	
1.	Profit (P) = SP – CP
2.	Loss (L) = CP – SP
3.	$P\% = \frac{P}{CP} \times 100$, $L\% = \frac{L}{CP} \times 100$
4.	$SP = MP - D$ or $SP = MP \times \frac{(100 - D\%)}{100}$
5.	$SP = CP \times \frac{(100 + P\%)}{100}$ or $SP = CP \times \frac{(100 - L\%)}{100}$
6.	$\text{Margin}\% = \frac{\text{Profit or Loss}}{SP} \times 100$
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- I. If $CP = 90$, $Gain\% = 10\%$, then $SP = ?$
- II. If $CP = 20$, $Loss\% = 25\%$, then $SP = ?$

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- I. If $CP = \text{Rs. } 5625$, $Gain = 20\%$, then $SP = ?$
- II. If $CP = \text{Rs. } 8040$, $Loss = 5\%$, then $SP = ?$

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I. If SP = Rs. 4060, Gain = 16%, then CP = ?

II. If SP = Rs. 5170, Loss = 12%, then CP = ?

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A man buys an article for Rs. 2750 and sells it for Rs. 2860. Find his gain per cent.

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If a radio is purchased for Rs. 490 and sold for Rs. 441, find the loss per cent.

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One gains 70 paise on Rs. 70. His gain percent is

Answer Options: 0.1%, 1%, 7%, 10%

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In terms of percentage profit, which is the best transaction?

CP (in Rs.)	Profit (in Rs.)
(1) 36	17
(2) 50	24
(3) 40	19
(4) 60	29

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A person incurs 5% loss by selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit?

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A book was sold for Rs. 2750 with a profit of 10%. If it were sold for Rs. 2575, then what would have been the percentage of profit or loss?

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A grocer purchased 80 kg of sugar at Rs. 13.50 per kg and mixed it with 120 kg sugar at Rs. 16 per kg. At what rate should he sell the mixture to gain 16%?

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The cost price of 25 articles is equal to the selling price of 20 articles. Find the Profit %

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By selling 66 meters of cloth, a person gains the cost price of 22 meters. Find the gain percent.

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A man purchases 8 pens for Rs. 9 and sells 9 pen for Rs. 8. How much profit or loss does he make

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A boy buys oranges at 9 for Rs. 16 and sells them at 11 for Rs. 20. What does he gain or lose (in percent)?

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If the profit percentage is 6.66%, find the margin percentage

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If the margin percentage is - 37.5%, find the loss percentage

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Thanks!

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Simple Interest –Terminologies

Principal (P) - Sum which is landed or borrowed

Rate of interest (R) -

Rate Percent of interest (R%) -

Interest - Simple (SI) or Compounded (CI)

Amount = Principal + Interest

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Simple Interest – Formula

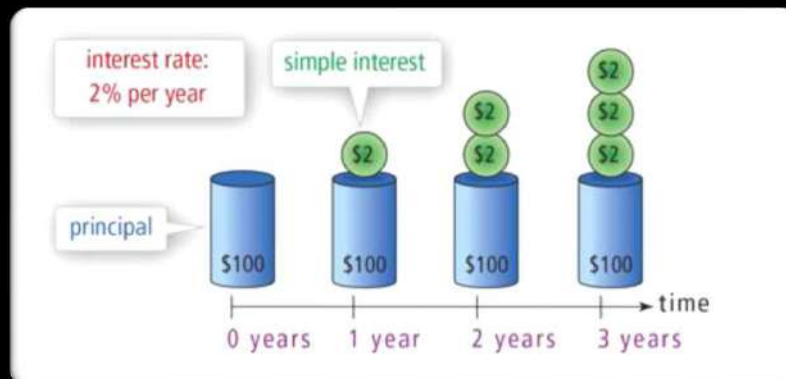
What is interest – Rent on Borrowed Money

Simple Interest =
$$SI = P \times \frac{R}{100} \times T$$

Note: The units of Rate of Interest per time period and Time period should match

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Simple Interest – Useful Results



- SI is same for each individual year.
- Amount on SI generate AP

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Simple Interest – Practice Exercise

1.	Principal (Rs.)	Rate (%)	Time	SI (Rs.)	Amount (Rs.)
i	240	8%	3 years		
ii	4000		4 years		5200
iii		10%	3 years		910
iv	840	8.33%	2 years		
v	72		5 months	3.75	
2.	Find out the Rate of interest (R) / Time Period (T) in following cases:				
i	A sum of money at SI grows to 8 times in 5 years.				R = ____
ii	A sum of money at SI grows to 15 times in 35 years.				R = ____
iii	A sum of money at SI grows to 8 times at 5% p.a.				T = ____
iv	A sum of money at SI grows to 12 times at 20% p.a.				T = ____

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Find SI on 700 for 3 years if rate of interest is 2% p.a.

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A certain sum of money amounts to $7/4$ times of itself in 3 years at simple interest.
Find the rate of interest

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A person borrows \$50000 for 5 years. What is the rate of simple interest charged if the person had to pay \$66000?

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After how many years would a sum doubles itself at 10% rate of simple interest?

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At a certain rate of simple interest a sum becomes three times in 15 years. In how many years will the sum become 9 times?

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If \$460 amounts to \$640 in 6 years at simple interest, what will it amounts to in 2 years at the same rate?

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A certain sum was put at certain rate at SI for 3 years. Had it been put at 2% points higher rate, it would have fetched \$360 more. Find the sum?

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A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is – Answer Choice - Rs. 650, Rs. 690, Rs. 698, Rs. 700

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Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B? **Answer Choices** - Rs. 6400, Rs. 6500, Rs. 7200, Rs. 7500

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Compound Interest

The Compound Interest Formula

Amount

Interest Rate (decimal)

$$A = P \left(1 + \frac{r}{n}\right)^{nt}$$

Time

Principal

Number of times interest
is compounded per unit 't'

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Simple Interest vs Compound Interest

SIMPLE INTEREST	COMPOUND INTEREST
Simple interest is interest charged on borrowed amount or interest on invested amount for entire period (it does not include interest on interest)	Compounded interest is interest calculated on the amount that includes the principal plus accumulated interest of the previous period.
Formula : Simple Interest = $(P \times I \times T) / 100$ Where; P = Principle Amount, I = Interest Rate; T = Time Period	Formula : Compound Interest = $P [(1+i)^t - 1]$ Where; P = Principle Amount, I = Interest Rate, T = Time Period
Example : A person borrows \$ 12,500 from a lender for three years at a rate of 10%. The simple interest calculation will be as below: Simple Interest = $(12,500 \times 10 \times 3) / 100$ = \$3,750	Example : A person borrows \$50,000 loan from Nainital Bank at a rate of 10% for 5 years compounded yearly. Principal = \$50,000; i = 0.10; t = 5 years. So, compound interest will be: CI = $50,000[(1+0.10)^5 - 1]$ = \$30,525.5

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Simple Interest vs Compound Interest

P = Rs. 4000, R = 10%

Year	Simple Interest	Compound Interest
1	400	400
2	400 (Total 800)	400 + 40 = 440
3	400 (Total 1200)	00 + 40 + 44 = 484
4	400 (Total 1600)	400 + 40 + 44 + 48.4 = 532.4

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Simple Interest vs Compound Interest

- Compound Interest is interest on interest
- Simple interest is same for every year
- For the first year SI = CI
- $\frac{\text{Amount after } (n+k)^{\text{th}} \text{ year}}{\text{Amount after } n^{\text{th}} \text{ year}} = \text{MF}^k$
- $\frac{\text{CI in } (n+k)^{\text{th}} \text{ year}}{\text{CI in } n^{\text{th}} \text{ year}} = \text{MF}^k$
- Difference between SI and CI for 2 years = $\frac{PR^2}{100^2}$
- Difference between SI and CI for 2 years = $\frac{PR^2}{100^2} \left(3 + \frac{R}{100} \right)$

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A bank charges a rate of 10% compounded annually. What is the total amount to be paid on a loan of \$36000 for 2 years?

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A certain sum becomes 1.728 times in 3 years under CI. Find R?

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If the rate of interest is 12.5%, then find the ratio of CI earned in 24th to 25th year ?

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If the rate of interest is 15%, then find the ratio of the total CI earned in the first 2 years to total SI earned in the first two years, if the Principal is same?

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A sum triples in 5 years at CI. Find in how many years the sum becomes 9 times of itself ?

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The difference between CI and SI for the first 2 years is 64. SI for the first two years is 640 then find P & R if both are the same for SI as well as CI.

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If a bank offers two schemes

- I. Semi annual compounding at 40% per annum.
- II. Quarterly compounding at 40% per annum.

Which of the two is a better scheme for depositors?

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Thanks!

SESSION 04 – Arithmetic II

SESSION 04 – Arithemtic II

by PANKAJ BURHANPURKAR



SESSION 04 – Arithemtic II

QCloud

A alone required 15 days to complete a given job. B alone can complete the same job in 10 days. If A and B are working together, how much time will they take to complete the same job?

SESSION 04 – Arithemtic II

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Ishan can do a certain job in 18 days. He works at it for 10 days and then Bharat alone completes it in 4 days. In how many days will Ishan and Bharat work together, to finish the work?

SESSION 04—Set II
Thematic II

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In a company, there are 500 employees. 180 employees do not have a car. 270 employees do not have a bike and 280 employees do not have a scooter, 10 employees do not have any of three. 420 employees have a car or a bike. 430 employees have a car or a scooter. 370 employees have a bike or a scooter.
How many employees have both a car and a scooter?
How many employees have all the three?
How many employees have exactly one type of vehicle?

SESSION 04—Set II
Thematic II

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Parul is twice as efficient as Parag. If both working together complete a work in 10 days, how many days would each have taken individually?

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One man can build a bridge in 100 days. One woman can build the same bridge in 120 days. One child can destroy the bridge completely in 200 days. If two men, three women and two children are working simultaneously on the bridge, in how many days will it be complete?

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Pipe A running alone can fill a cistern in 10 hours. Pipe B running alone can fill a cistern in 12 hours. In how many hours will both running together fill the cistern?

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If a factory can manufacture b computer screens in n days at a cost of c dollars per screen, then which of the following represents the total cost, in dollars, of the computer screens that can be manufactured, at that rate, in m days?
 bcm/n , bmn/c , mc/bn , bc/mn

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A man and two boys can do a piece of work in 10 days. Two men and a boy can do the same work in 8 days. In how many days can a man and a boy together complete the work?

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Thanks!

SESSION 04 – Arithmetic II

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A cyclist covers a distance of 750 m in 2 min 30 sec. What is the speed in kmph of the cyclist?

Which of the following trains is the fastest?

(a) 25 m/s; (b) 1500 m/min; (c) 90 kmph

A man in a train notices that he can count 21 telephone posts in one minute. If they are known to be 50 m apart, then at what speed is the train travelling?

If a man walks at the rate of 5 kmph, he misses a train by 7 minutes. However, If he walks at the rate of 6 kmph, he reaches the station 5 minutes before the arrival of the train. Find the distance covered by him to reach the station.

How many minutes does Suman take to cover a distance of 400 m if she runs at a speed of 20 kmph?

While covering a distance of 24 km, a man noticed that after walking for 1 hour and 40 minutes, he covered $\frac{5}{7}$ of the remaining distance. What was his speed in m/s?

From home to the office, if I travel at $\frac{3}{5}$ of my usual speed, I am late by 12 minutes. Find the time that I take usually and the time taken at the reduced speed.

Two trains start simultaneously, one from Bombay to Kolkata and the other from Kolkata to Bombay. They meet each other at Nagpur which is at a distance of 700 km from Bombay. If the distance between Bombay and Kolkata is 1600 km, find the ratio of their speeds.

Two friends start walking towards each other with speeds in the ratio 3 : 4. When they meet it is found that the faster of them has covered 25 m more than the slower. Find the distance that separated them initially if they are walking opposite directions.

A Police-man starts chasing a thief. The ratio of the speeds of the thief and the policeman is in the ratio 9:11 and when the policeman catches the thief it is found that the policeman has covered 60 meters more than the thief. How much distance did the Police have to run to nab the thief?

In the movie Ghulam, Aamir is able to spot the approaching train when it is 2 km away. He has to run towards the train and reach the red kerchief hung on a pole 400 meters away from him before the train reaches the pole. How fast must Aamir run if the speed of the train is 36 kmph so that he just manages to reach the kerchief at the same time as the train reaches it?

A thief is spotted by a policeman from a distance of 100 m. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8 kmph and that of the policeman is 10 kmph, what is the distance that the thief would have covered before he is caught?

A train is travelling at a steady speed of 75 kmph. Currently, the distance between the train and a car coming from the opposite direction towards the train is 250 km. What should be the speed of the car so that the two meet in 1 hr 15 mins?

A person covers 12 km in 15 hrs

Column A
His speed in m/s

Column B
0.4

X covers first half the distance at 15 m/s and the remaining 300 m in 20 sec.

Column A

His average speed for
entire journey

Column B

15 mps

In 200 m race A can give B a startup of 10 m and C a startup of 15 m

Column A

Startup B can give
C in the same race

Column B

5 m



Thanks!

Class Practice Exercise 15

Profit, Loss, Discount I

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. A man buys an article for Rs. 2750 and sells it for Rs. 2860. Find his gain percent.
 (1) 2.5% (2) 3.2%
 (3) 4% (4) 4.4%
2. Prakash used to sell ready made shirts from his stall in the local market. He purchased 160 shirts from a company outlet at Rs. x per shirt. If he sells 20% of the shirts at 15% profit and 37.5% of the remaining shirts at 40% profit, then at what percentage profit should he sell the remaining to make a profit of 25% on the whole?
 (1) 6.25% (2) 25%
 (3) 20% (4) 325%
3. The Santabil shop offers that get a T-shirt free on the purchase of every three T-shirts. But they mark-up the price of the T-shirts by 60%. If a customer buys three T-shirts, then what is the profit or loss percentage for the store?
 (1) 12.5% profit (2) 20% profit
 (3) 25% profit (4) 16.67% loss
4. By selling an article at $\frac{2}{5}$ th of the marked price, there is a loss of 25%. The ratio of the marked price and the cost price of the article is
 (1) 8 : 15 (2) 8 : 5
 (3) 15 : 8 (4) Data Inadequate
5. Two varieties of rice A and B are mixed in the ratio of 3:5 by weight. By selling the mixture at the price of the dearer variety A, a profit of 20% is made. If the price of the dearer variety is Rs. 30 per kg then find the price of the cheaper variety B (per kg).
 (1) Rs. 22 (2) Rs. 20
 (3) Rs. 18 (4) Rs. 24
6. A shopkeeper marks an item for sale at Rs. 850 and then gives a discount of 20% on it. The item cost the shopkeeper Rs. 500 when he purchased the same from the wholesale. What is the profit (in percentage) earned by the shopkeeper?
 (1) 28% (2) 32%
 (3) 36% (4) 40%
7. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is –
 (1) 32/7% (2) 60/11%
 (3) 10% (4) 12%
8. A man buys shares at a discount of Rs. x. Later he sold all but 10 of the shares he purchased at a premium of Rs. x. If his investment was Rs. 4,500 and proceeds from the sale were Rs. 6,250, then how many shares did he buy originally? [Assume the face value of shares as Rs. 100.]
 (1) 40 (2) 50
 (3) 60 (4) 75
9. Avinash buys a certain number of apples at 3 for Rs. 16 and sells them at 5 for Rs. 28. Find his profit/loss percentage in the transaction.
 (1) 4% loss (2) 4% gain
 (3) 5% gain (4) 5% loss
10. If the profit on selling 25 mangoes is the cost price of 15 mangoes, then what is the profit percentage?
 (1) 30% (2) 40%
 (3) 50% (4) 60%
11. A dealer sells a set of furniture for Rs.900, which is 80% more than the price he paid for it. At what price must he sell the same set to earn a profit of 120% on the cost price?
 (1) Rs.1,000 (2) Rs.1,100
 (3) Rs.1,200 (4) Rs.1,250
12. Three partners A, B and C run a business. The ratio of their shares in the annual profit is 3 : 2 : 6. If A's capital is less than B's capital and more than C's capital, whose capital is invested for the maximum period?
 (1) A (2) C
 (3) B (4) Can't determine
13. The cost of manufacturing a commodity increased by 20%. A trader who did not revise the selling price noted that there was a drop of Rs.20 in his profit. What was the original cost price (in Rs.)?
 (1) Rs. 80 (2) Rs. 90
 (3) Rs. 100 (4) Rs. 120
14. If a radio is purchased for Rs. 490 and sold for Rs. 441, find loss percent.
 (1) 8% loss (2) 10% loss
 (3) 12% loss (4) 15% loss
15. A shopkeeper buys 10 apples at a certain price. But he steals three more from the dealer. Assuming that one apple was rotten and he sells the rest at the cost price, what is his profit?
 (1) 12% (2) 20%
 (3) 25% (4) 18%
16. If cost price of 4 articles is equal to selling price of 5 articles, then find the profit/loss percentage.
 (1) 20% loss (2) 25% profit
 (3) 25% loss (4) None of these
17. In the above problem, the shopkeeper quotes a price that is 10% more than the cost price, but one more apple gets rotten. What's his new profit?
 (1) 12.1% (2) 21%
 (3) 18% (4) 20%

18. The cost price of a commodity is Rs.1,331. By selling it at a discount of Rs.100, the dealer makes a profit of 9.09%. If he decides not to give any discount, what will be the profit earned by him?
 (1) Rs.131 (2) Rs.121
 (3) Rs.221 (4) Rs.231
19. Even if an article is sold at only $\frac{3}{5}$ th of the actual selling price, there is a profit of 20%. The profit on selling 45 such articles at the actual selling price is equal to the
 (1) cost price of 45 articles
 (2) selling price of 45 articles
 (3) cost price of 30 articles
 (4) selling price of 30 articles
20. A man sells an article at a profit of 20%. If he had bought it at 20% less and sold it for Rs. 5 less, he would have gained 25%. Find the cost price (in Rs.) of the article.
 (1) Rs. 20 (2) Rs. 25
 (3) Rs. 30 (4) Rs. 35
21. A student gets 60% of the maximum marks. If he gets 50% more than the pass mark, then the pass mark is what percentage of the maximum marks?
 (1) 75% (2) 37.5%
 (3) 40% (4) 50%
22. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is –
 (1) 15 (2) 16
 (3) 18 (4) 25
23. Pramod had sufficient money to purchase 12 mangoes or 6 apples or 20 bananas. He retained 10% of the money for his transport charges. He spent the entire remaining money in purchasing fruits. If he purchased exactly 2 apples and at least one mango, then find the maximum number of bananas he could have purchased.
 (1) 5 (2) 4
 (3) 8 (4) 10
24. After allowing a discount of 11.11%, a trader still makes a gain of 14.28%. At how much percentage above the cost price does he mark his goods?
 (1) 28.57% (2) 35%
 (3) 22.22% (4) 25.18%
25. An article, which was bought by C from B, is sold by C for Rs. 3000. B sold the article to C at a profit which is the same as the profit (in Rs.) made by A when A sold it to B. The magnitudes of the percentage of the profit made by B and the percentage of loss suffered by C are the same. If the gain of A is 33.33%, at what price did A purchase it?
 (1) Rs. 2400 (2) Rs. 3200
 (3) Rs. 4000 (4) None of these
26. Find SP when CP – Rs. 5625, Gain = 20%
 (1) Rs. 6600 (2) Rs. 6725
 (3) Rs. 6750 (4) Rs. 6800
27. A man buys oranges at the rate of Rs.2 for 6 and sells the whole lot at the rate of Rs.3 for 7. In this process, he gained Rs.20. How many oranges did he purchase?
 (1) 420 (2) 210
 (3) 360 (4) 250
29. If selling price is doubled, the profit triples. Find the profit percent.
 (1) 200/3% (2) 100%
 (3) 316/3% (4) 120%
30. Hrithik and Kartik work for a company and they get a monthly salary of Rs. 36,000 each. After a year of working, Hrithik gets a hike of 25% in his salary and Kartik gets a hike of 33-33%. After the hike, Kartik's expenditure is 50% of that of Hrithik's and 25% of his own salary. What percentage of Hrithik's savings are Kartik's savings after the hike?
 (1) 25% (2) 28.6%
 (3) 100% (4) 171.4%
31. Mr. Alphonso buys 2 lots of mangoes. The first lot costs him Rs.30 per dozen and the second lot Rs.50 per dozen. He sold all the mangoes at Rs.40 per dozen. Find his profit or loss percentage if he spent the same amount of money on each of the lots.
 (1) $6\frac{2}{3}\%$ loss (2) $6\frac{2}{3}\%$ profit
 (3) 5% profit (4) 5% loss
32. A, B and C start a venture with different amounts. A starts with 11000 and increases his investment by Rs. 1,000 every month. B starts with Rs. 2,000 and increases his investment by Rs. 2000 every 2 months. C starts with Rs. 3000 and increases his investment by Rs. 3000 every 3 months. Assuming that the amount is invested for one year, the ratio in which A, B and C share the annual profit is
 (1) 33 : 14 : 13 (2) 13 : 14 : 15
 (3) 14 : 13 : 15 (4) 15 : 13 : 14
33. Find SP when CP – Rs. 8040, Loss = 5%
 (1) ₹ 7600 (2) ₹ 7638
 (3) ₹ 7700 (4) ₹ 7500
34. A, B, C and D have equal amounts with them. The following transactions of money take place among them the following sequence. A gives D 10% of what A has. B gives A 50% of what B has. C gives 30% of what C has. D gives B 25% of what D has. C gives A 30% of what C has. D gives B 20% of what D has. After the above transactions take place, which of the following represents the descending order of the amounts they have?
 (1) ABCD (2) BADC
 (3) ABDC (4) None of these
35. By selling toffees at 56 a rupee, a vendor loses 30%. How many toffees for a rupee should he sell in order to gain 40%?
 (1) 16 (2) 24
 (3) 28 (4) 30

36. A shop keeper hiked the price of an article by $x\%$ and then gave discount and the price of the article decreased by Rs. 2000. Once again, the price of the article was increased by $\frac{x}{2}\%$ and then decreased by $\frac{x}{2}\%$ and it was finally sold at Rs. 47520. What was the initial price of the article? (In Rs.)
 (1) 48,500 (2) 50,000
 (3) 51,000 (4) 46,000
37. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
 (1) 30% (2) 70%
 (3) 100% (4) 250%
38. A retail fruit vendor purchases two varieties of oranges at the rates of 12 oranges for Rs. 18 and 18 oranges for Rs. 12. He mixes the two varieties in the ratio 2 : 3 and sells the mixed stock at a price of Rs. 144 for to dozens. What percentage of profit or loss does the he make?
 (1) $16\frac{2}{3}\%$ (2) 20%
 (3) $33\frac{1}{3}\%$ (4) 25%
39. The cost of manufacturing a commodity increased by 20%. A trader who did not revise the selling price noted that there was a drop of Rs.20 in his profit. What is the original cost price?
 (1) Rs.100 (2) Rs.200
 (3) Rs.500 (4) Rs.150
40. John and Algore were the only two contestants for presidency. The number of votes for Algore was 50% of the votes for John. If 1000 voters who voted for Algore, had instead voted for John, than the votes for Algore would have been only 25% of the votes for John. Assuming all the voters cast their votes in favour of either John or Algore (but not both), how many votes were polled?
 (1) 2500 (2) 5000
 (3) 10500 (4) 7500
41. Find CP when SP = Rs. 4060, Gain = 16%
 (1) Rs. 3400 (2) Rs. 3500
 (3) Rs. 3600 (4) Rs. 3700
42. A shopkeeper sells 120 articles each at a price of Rs. 250. To the persons who do not bargain he does not give any discount and makes 50% profit on each article. To those who bargain, he gives a discount of 20%. If one in four customers do not bargain. Find his profit percentage on selling all the 120 articles.
 (1) 25% (2) 30%
 (3) 375% (4) 27.5%
43. A rifle and a pistol were sold for Rs.564, making a profit of 25% on the rifle and a profit of 20% on the pistol. By selling the rifle and the pistol for Rs.563, the profit would be 20% on the rifle and 25% on the pistol. Find the ratio of difference and the sum of the cost prices of the rifle and the pistol.
 (1) $\frac{1}{11}$ (2) $\frac{1}{19}$
 (3) $\frac{1}{23}$ (4) $\frac{1}{25}$
44. If 20% of the goods are sold at 50% profit, 40% of the goods at 20% loss. 20% of the goods at 5% loss and the remaining at no loss or no gain, then find the overall profit percentage.
 (1) 8% (2) 0.5%
 (3) 4% (4) 1%
45. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?
 (1) 3 (2) 4
 (3) 5 (4) 6
46. If the discount percentage of an article is equal to the profit percentage, and the ratio of the cost price and the marked price is 8:13, then find the cost price of the article if its selling price is Rs. 120.
 (1) Rs. 50 (2) Rs. 76
 (3) Rs. 97 (4) Rs. 1105
47. John goes to Archie's gallery to purchase a gift for his girlfriend Bipasha for Valentine's Day. John has the privilege of buying the gifts at a 24% discount on the listed price. Bipasha ditches John on Valentine's Day, so he decides to sell the purchased gift to someone else to avoid any more losses. Suppose he wants to make a profit of 25% after allowing a discount of 20%, by what percent should his marked price be greater than the original listed price?
 (1) 17% (2) 17.75%
 (3) 18.75% (4) 19%
48. Natwar, the proprietor of a book shop allows the following percentages of successive discounts on four identical dictionaries. Which of the following sequences of successive discounts, gives him the greatest profit?
 (1) 5%, 10%, 10% (2) 1%, 12%, 12%
 (3) 4%, 15%, 6% (4) 9%, 9%, 7%
49. Find CP when SP = Rs. 5170, Loss = 12%
 (1) Rs. 5800 (2) Rs. 5875
 (3) Rs. 5950 (4) Rs. 6000
50. Ankush bought a plot of land for Rs. 50,000. After two years he found that the market value of land increased by 300%. At this state, he has two options. He can sell at the market rate or invest some more money on marketing and than get a better price. He estimated that if he spent Rs. 10,000 on marketing, he would get 25% more than the market value. Find the difference between the profit percentages in the two options (in percentage points).
 (1) 10 (2) $12\frac{1}{2}\%$
 (3) $14\frac{2}{7}\%$ (4) $16\frac{2}{3}\%$
51. Costs of (2a, 3m, 4o) are Rs. 6 more than (1a, 2m, 2o). Costs of (1a, 2m, 1o) are Rs. 8 less than (3a, 3m, 5o). If the list price of (3a, 1m, 6o) is Rs. 12, what discount percentage ensures a 5% net profit?
 (1) 10% (2) 7.5%
 (3) 15% (4) 12.5%

52. A and B invest a total of Rs. 3,00,000 in a venture and record a profit of Rs.75,000. As per the agreement B collects 20% of the profit as his salary for handling the business and then they share the remaining profit in the ratio of their investment. If they share the entire profit in the ratio of their investments, A would have got Rs. 7,500 more than what he had got as per the agreement. A's investment is
 (1) Rs. 31,50,000 (2) Rs. 1,45,000
 (3) Rs. 1,75,000 (4) Rs. 1,25,000
53. The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling the same article for Rs. 1280. At what price should the article be sold to make 25% profit?
 (1) Rs. 2000 (2) Rs. 2200
 (3) Rs. 2400 (4) Data Inadequate
54. The population of a country increases by 10% every year, and the national income increases by 21% every year. What is the percentage change in the per capita income of the country every year?
 Per capita income = National Income/Population
 (1) 10% (2) 11%
 (3) 2.1% (4) 20%
55. If the cost price of 6 articles is the same as the selling price of 5 articles, the profit percentage will be
 (1) 25% (2) $22\frac{1}{2}\%$
 (3) 20% (4) $18\frac{3}{4}\%$
56. A trader sold an article at a profit of $9\frac{1}{11}\%$. Had the article been bought for Rs. 1300 less and sold for Rs. 3000 less, the loss would be $9\frac{1}{11}\%$. What is the original cost price of the article?
 (1) Rs. 9500 (2) Rs. 9000
 (3) Rs. 5000 (4) Rs. 7700
57. One gains 70 paise on Rs. 70. My gain percent is
 (1) 0.1% (2) 1%
 (3) 7% (4) 10%
58. A milk vendor who sells milk at Rs. 10.50 per half a litre, finds a milk can lying by the roadside unclaimed and unattended. He picks up the milk can and tests the milk. He finds it fresh and of the same quality as the milk he already has with him. He initially has 52 litres of milk, while the can that he finds has 13 litres. He mixes the two quantities and sells at a reduced price but maintains his percentage of profit. By what proportion does he decrease the price per litre?
 (1) $(1/4)$ th (2) $(1/5)$ th
 (3) $(1/6)$ th (4) $(1/3)$ rd
59. A shopkeeper bought some pencils at 2 for Rs. 1, and an equal number at 3 for Rs. 2. He sold the entire lot at 5 for Rs. 3. Find out his gain or loss percentage.
 (1) $2\frac{6}{7}\%$ loss (2) $3\frac{6}{7}\%$ gain
 (3) $2\frac{5}{7}\%$ gain (4) $3\frac{5}{7}\%$ loss
60. Which of the following options is the best investment?
 (1) 2% stock at Rs. 75 (2) 3% stock at Rs. 150
 (3) 3% stock at Rs. 100 (4) 4% stock at Rs. 175
61. A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?
 (1) Rs. 18.20 (2) Rs. 70
 (3) Rs. 72 (4) Rs. 88.25
62. I bought shares at DPL for Rs. 32400 at 3108 in 2004, in 2006. I sold all the shares quoting at Rs. 117. But the next day the sensex crashed and the value of each share at DPL was quoted at Rs. 80. If I buy the same number of DPL shares using the money that I got by selling the shares, then approximately what percent of the money I get will I be left with me? (DPL has not paid any dividend during this period).
 (1) 46% (2) 47%
 (3) 52% (4) 32%
63. The spring balance of a trader showed 800 gm for 1kg. What is the net result for the shopkeeper, is it a profit or a loss and by what percent?
 (1) 25% loss (2) 20% profit
 (3) 20% loss (4) 25% profit
64. The marked price of an article is 80% more than the cost price. When the percentage of discount is increased from 22.5% to 37.5%. The profit would be Rs. 24 less. What is the marked price at the article?
 (1) Rs. 160 (2) Rs. 180
 (3) Rs. 216 (4) Rs. 236
65. In terms of percentage profit, which is the best transaction?
- | | CP (in Rs.) | Profit (in Rs.) |
|-----|-------------|-----------------|
| (1) | 36 | 17 |
| (2) | 50 | 24 |
| (3) | 40 | 19 |
| (4) | 60 | 29 |
66. A trader underweighs rice and gets a profit of 40%. Had he used the correct weight he would have still got a profit of 26%. What weight does he use for 1 kg?
 (1) 860 grams (2) 900 grams
 (3) 940 grams (4) 820 grams
67. A man bought a shirt at $3/4$ th of its list price and sold it at 50% above its list price. What is his percentage gain in the transaction?
 (1) 50% (2) 75%
 (3) 25% (4) 100%
68. The cost of producing a saree is (800 and it is sold at a profit of 25%. The components of the cost (as a percentage of the cost of production per saree) are 5% on cotton. 5% on dyes. 50% on spinning. 20% on marketing and 20% on packing and delivery. If the cost of dyes increases by 50%, marketing cost by 25% and the selling price is decreased by 10%, and all the other components remain the same, then what is the percentage decrease in the profit?
 (1) 84% (2) 20%
 (3) 81.6% (4) 80%

69. A person incurs 5% loss by selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit?
 (1) 1260 (2) 1200
 (3) 1280 (4) 1254
70. In 2005, a company decided to distribute 80% of its profits among all its employees and spend 10% of the profit for further investment. The remaining profit is taken by the chairman of the company. In 2006, the profit doubled and the company distributed 50% more among the employees than in the previous year. Find the percentage increase in the share of the chairman, if the company reinvested 50% of the profit remaining after the employees were given their share.
 (1) $333\frac{1}{3}\%$ (2) 250%
 (3) 300% (4) 400%
71. A trader bought 10 kg oranges for Rs. 405, of which 1 kg was found rotten. If he wishes to make a profit of 10%, at what rate per kg should he sell the remaining oranges?
 (1) Rs. 44.55 (2) Rs. 48.66
 (3) Rs. 51.25 (4) Rs. 49.50
72. In 2004, 40% of the sales of the product A is equal to 60% of the sales of product B. If the total sales of A and B in 2004 (in lakhs of rupees) is 9000 and the sales of A and B increased by 10% and 20% respectively in 2005, then the difference in the sales of A and B in 2005 (in lakhs of rupees) is
 (1) 2520 (2) 1860
 (3) 1620 (4) 2120
73. A book was sold for Rs. 2750 with a profit of 10%. If it were sold for Rs. 2575, then what would have been the percentage of profit or loss?
 (1) 3% profit (2) 5% loss
 (3) 2% profit (4) 3% loss
74. Chetan started a cable TV service with an investment of Rs. 2,00,000. After a few months David joined him with an investment of Rs. 1,50,000. If at the end of the year David's share was Rs. 3,00,000 out of a total profit of Rs. 7,80,000 after how many months did David join Chetan?
 (1) 2 (2) 4
 (3) 8 (4) 10
75. A vendor bought bananas at 6 for Rs. 10 and sold them at 4 for Rs. 6. Find his gain or loss percent.
 (1) 10% loss (2) 10% profit
 (3) 11.11% loss (4) 20% loss
76. A trader, by announcing a discount of 2% on an article reduces his profit from Rs. 600 to Rs. 500. Find the cost price of the article.
 (1) Rs. 3,500 (2) Rs. 3,800
 (3) Rs. 4,400 (4) Rs. 4,800
77. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
 (1) Rs. 1090 (2) Rs. 1160
 (3) Rs. 1190 (4) Rs. 1202
78. What percentage of profit is made by selling an article at a certain price, if by selling it at half the original price there would be a loss of 20%?
 (1) 60% (2) 20%
 (3) 40% (4) 30%
79. A man bought toffees at 3 for a rupee. How many for a rupee must he sell to gain 50%.
 (1) 2 (2) 4
 (3) 1.5 (4) 5
80. Mangoes bought, at the rate of 14 for Rs. 50 are sold at a profit of 40%. How many mangoes will be sold for Rs. 100?
 (1) 24 (2) 25
 (3) 20 (4) 18

Class Practice Exercise 16

Profit, Loss, Discount II

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. If the cost price is 96% of the selling price, then what is the profit percent?
 (1) 4.16% (2) 4%
 (3) 4.5% (4) 4.25%
2. A and B start a business by investing Rs. 25,000 and 20,000 respectively. A invests his capital only for 6 months while B invests for $\frac{5}{6}$ th of the period for which A invests. 5% of the profit is given as charity. Find the total profit, if A's share is Rs. 11,400.
 (1) Rs. 16,000 (2) Rs. 30,000
 (3) Rs. 45,000 (4) Rs. 20,000
3. A grocer purchased 80 kg of sugar at Rs. 13.50 per kg and mixed it with 120 kg sugar at Rs. 16 per kg. At what rate should he sell the mixture to gain 16%?
 (1) Rs. 17.40 (2) Rs. 16.50
 (3) Rs. 18 (4) Rs. 15
4. A man sold a Doberman and a German Shepherd for Rs. 5,750 each. On the Doberman he made a profit of 25%, and on the German Shepherd he made a profit of 15%. Find the approximate profit percentage on the whole transaction.
 (1) 23% (2) 20%
 (3) 22% (4) 18%
5. The CP of 21 article is equal to SP of 18 articles. Find the gain or loss percent.
 (1) 16.66% gain (2) 16.66% loss
 (3) 14.28% gain (4) 20% gain
6. Selling an item at $\frac{5}{6}$ of its marked price results in a loss of 10%. What is the percentage of profit/loss, if a discount of 5% is offered on the marked price?
 (1) 3% loss (2) 2.6% profit
 (3) 2.6% less (4) 3% profit
7. Pure ghee costs Rs. 100 per kg. After adulterating it with vegetable oil costing Rs. 50 per kg, a shopkeeper sells the mixture at the rate of Rs. 96 per kg, thereby making a profit of 20%. In what ratio does he mix the two?
 (1) 3 : 2 (2) 2 : 3
 (3) 4 : 5 (4) 1 : 1
8. A solution of sugar and water contains 20% sugar by weight. From this solution, if 25 kg of water is evaporated, the mixture now contains 25% of sugar. Find the quantity of sugar in the solution initially.
 (1) 50 kg (2) 125 kg
 (3) 100 kg (4) 25 kg
9. Sam purchased 20 dozen of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit?
 (1) 3.5% (2) 4.5%
 (3) 5.6% (4) 6.5%
10. A man sold his two scooters, LML and Chetak for a total of Rs. 20,000 making a profit of 20% on Chetak and a profit of 10% on LML. If the profit percentages are interchanged, then the total amount he receives on the sale is Rs. 900 less. Find the difference in the cost prices of Chetak and LML.
 (1) Rs. 4,000 (2) Rs. 7,000
 (3) Rs. 5,500 (4) Rs. 9,000
11. A dishonest dealer professes to sell his goods at cost price but uses a weight of 960 gms for a kg weight. Find his gain percent.
 (1) 4.66% (2) 4%
 (3) 4.5% (4) 4.2%
12. If a retailer makes a profit of $16\frac{2}{3}\%$ on his selling price, what is his profit percentage on the cost price?
 (1) $16\frac{2}{3}\%$ (2) 10%
 (3) 20% (4) None of these
13. By selling 33 meters of cloth, one gains the selling price of 11 meters. Find the gain percent.
 (1) 50% (2) 33.33%
 (3) 25% (4) 40%
14. A paper merchant overweighs paper by 30% while buying it at Rs. 5 per kg and then sells it to a recycling plant at Rs. 8 per kg. What is his profit percentage?
 (1) 8% (2) 54%
 (3) 108% (4) None of these
15. If the manufacturer gains 10%, the wholesale dealer 15% and the retailer 25%, then find the cost of production of a table, the retail price of which is Rs. 1265?
 (1) 800 (2) 900
 (3) 973 (4) 1012
16. I bought a bundle of sheets of paper from the manufacturer at the rate of 10 sheets for Rs. 5. I sold these sheets at the rate of 20 sheets for Rs. 30. Find how many sheets were there in the bundle, if I made a profit of Rs. 500.
 (1) 500 (2) 1000
 (3) 5000 (4) 2500
17. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is –
 (1) 30% (2) 100/3%
 (3) 35% (4) 44%
18. A man sells an article at a profit of 25%. Had he bought it at 25% less and sold for Rs. 25 less, he would still have gained 25%. Find the cost of the article.
 (1) Rs. 80 (2) Rs. 40
 (3) Rs. 60 (4) Rs. 50

19. Monika purchased a pressure cooker at $9/10^{\text{th}}$ of its selling price and sold it at 8% more than its SP find her gain percent.
 (1) 20% (2) 18%
 (3) 16.66% (4) 25%
20. A, B and C invested capitals in the ratio 4 : 5 : 6. At the end of the year, they received the profits in the ratio 6 : 5 : 4. Find the ratio of time periods for which they invested their capitals.
 (1) 9 : 6 : 2 (2) 9 : 8 : 6
 (3) 9 : 6 : 4 (4) 9 : 3 : 4
21. A vendor bought bananas at 6 for Rs. 10 and sold them at 4 for Rs. 6. Find his gain or loss percent.
 (1) 10% loss (2) 10% profit
 (3) 11.11% loss (4) 20% loss
22. Amala and Bindu started a pet care service without any capital. After 6 months Amala had to leave and with her departure the firm's pet strength came down by 50%. Bindu continued till the end of the year. The firm gets a net profit (income - expenses) of Rs. 30,000 at the end of the year. What share of the profit will Amala get if they agreed to divide the profit proportionally to the pets brought in by each of them and the time for which they took care of the pets?
 (1) Rs. 10,000 (2) Rs. 15,000
 (3) Rs. 20,000 (4) Rs. 5,000
23. An article is sold at a certain price. By selling it at $2/3$ of that price one loses 10%. Find the gain percent at original price.
 (1) 30% (2) 35%
 (3) 25% (4) 40%
24. Mutual Fund XYZ offers three schemes where one can invest for a period of 30 days.
Scheme A: It invests the money in debt funds and gives a steady return of 1.2%
Scheme B: It takes a bearish view of the market and short sells it and offers a return of - 6% if the market goes up and +9% return if the market goes down
Scheme C: It takes a bullish view of the market and buys into it and gives a +13% returns if the market goes up and a return of - 7% if the market goes down. Find the maximum guaranteed return that any person can expect if he invested in one or more of these scheme.
 (1) 13/7% (2) 15/7%
 (3) 16/7% (4) 17/7%
25. On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is –
 (1) Rs. 45 (2) Rs. 50
 (3) Rs. 55 (4) Rs. 60
26. A man purchases a certain number of chocolates at 2 per rupee and the same number of pepper-mints at 5 per rupee. He mixes them together and sells them at 3 per rupee. What is his approximate profit or loss percentage?
 (1) 5% profit (2) 5% loss
 (3) 4% profit (4) 4% loss
27. A tradesman sold an article at a loss of 20%. If the selling price had been increased by Rs. 100, there would have been a gain of 5%. What was the cost price of the article?
 (1) Rs. 360 (2) Rs. 400
 (3) Rs. 425 (4) Rs. 450
28. What is the percentage change in the volume of a cylinder if its height increases by 20% and radius remains the same?
 (1) 20% increase (2) 10% increase
 (3) 20% decrease (4) 10% decrease
29. When a plot is sold for Rs. 18,700, the owner loses 15%. At what price must that plot be sold in order to gain 15%?
 (1) Rs. 21,000 (2) Rs. 22,500
 (3) Rs. 25,300 (4) Rs. 25,800
30. With the profit made by selling 25 pens, a trader is able to buy 50 pens. Find the profit percentage.
 (1) 50% (2) 100%
 (3) 200% (4) 300%
31. A man sells an article at a profit of 25%. If he had bought it at 20% less and sold it for Rs. 10.50 less, he would have gained 30%. Find the cost price of the article.
 (1) Rs. 40 (2) Rs. 50
 (3) Rs. 55 (4) Rs. 60
32. If a kerosene dealer sells kerosene at Rs. 8 per litre, he loses Rs. 400, but if he sells at Rs. 10 per litre he makes a profit of Rs. 400. How many litres of kerosene did the dealer sell?
 (1) 250 (2) 200
 (3) 350 (4) 400
33. A man bought toffees at 3 for a rupee. How many for a rupee must he sell to gain 50%?
 (1) 1.5 toffees per rupee
 (2) 2 toffees per rupee
 (3) 2.5 toffees per rupee
 (4) 3 toffees per rupee
34. If a retailer sells an item for $4/5^{\text{th}}$ of its list price, he realises a profit of 12%. During the annual clearance sale he wants to offer a huge discount to attract customers. What is the maximum discount he can offer without incurring a loss?
 (1) 20% (2) 25%
 (3) 40% (4) None of these
35. The price of a jewel, passing through three hands, rises on the whole by 65%. If the first and the second seller earned 20% and 25% profit respectively, find the percentage profit earned by the third seller.
 (1) 8% (2) 10%
 (3) 12% (4) 15%
36. A music system was sold at a loss of 10%. If it were sold for Rs. 700 more, there would have been a gain of 4%. Find the cost price of the music system.
 (1) Rs. 8,500 (2) Rs. 5,000
 (3) Rs. 7,000 (4) Rs. 6,000

37. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is -
 (1) 100/7% gain (2) 15% gain
 (3) 100/7% loss (4) 15% loss
38. A trader sells 30% of his stock at 14% loss. At what profit percentage should he sell the remaining stock to get an overall profit of 7%?
 (1) 14% (2) 16%
 (3) 18% (4) 20%
39. A man sold two flats for Rs. 6,75,958 each. On one he gains 16% while on the other he loses 16%. How much does he gain or loses in whole transaction?
 (1) 2.5% gain (2) 2.56% loss
 (3) 4% loss (4) 3% gain
40. By selling 50 metres of cloth, a merchant incurs a loss equal to the cost price of 10 metres. Find the loss percentage.
 (1) 25% (2) $33\frac{1}{3}\%$
 (3) 20% (4) $16\frac{2}{3}\%$
41. A grocer purchased 80 kg of sugar at Rs. 13.50 per kg and mixed it with 120 kg sugar at Rs. 16 per kg. At what rate should he sell the mixture to gain 16%?
 (1) ₹16.8 per kg (2) ₹17.0 per kg
 (3) ₹17.4 per kg (4) ₹18.0 per kg
42. Atul and Balram start a refrigeration service centre with an investment of Rs. 1,40,000 and Rs. 2,10,000 respectively. After 6 months Atul increases his capital by two-seventh and Balram by one-seventh. If Balram's share of the annual profit is Rs. 2,70,000. What was Atul's share? (in Rs.)
 (1) 1,92,000 (2) 2,40,000
 (3) 2,80,000 (4) 2,08,000
43. A dealer sold three-fourth of his articles at a gain of 20% and the remaining at cost price. Find the gain earned by him in the whole transaction.
 (1) 12% (2) 15%
 (3) 16% (4) 18%
44. By selling 60 metres of cloth, a merchant gains the selling price of 10 metres. The gain percentage is
 (1) 25% (2) 20%
 (3) 15% (4) 16%
45. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is -
 (1) 100/17% loss (2) 100/17% gain
 (3) 20/3% gain (4) None of these
46. A shopkeeper bought a table marked at Rs. 600 at successive discounts of 10% and 20% respectively. He spent Rs. 8 on transportation and sold the table for Rs. 450. Find his profit percentage.
 (1) $2\frac{7}{11}\%$ (2) $2\frac{3}{11}\%$
 (3) $2\frac{5}{11}\%$ (4) $2\frac{9}{11}\%$
47. A man bought a horse and a carriage for Rs. 3000. He sold the horse at a gain of 20% and the carriage at a loss of 10%, thereby gaining 2% on the whole. Find the cost of the horse.
 (1) Rs. 1000 (2) Rs. 1200
 (3) Rs. 1500 (4) Rs. 1800
48. A person sold his watch for Rs. 75 and got a profit percentage numerically equal to the cost price (in Rs.). The cost price of the watch is
 (1) Rs. 60 (2) Rs. 40
 (3) Rs. 50 (4) Rs. 75
49. Pure ghee costs Rs. 100 per kg. After adulterating it with vegetable oil costing Rs. 50 per kg, a shopkeeper sells the mixture at the rate of Rs. 96 per kg, thereby making a profit of 20%. In what ratio does he mix the two?
 (1) 3 : 2 (2) 2 : 3
 (3) 5 : 3 (4) 4 : 1
50. Nala and Bheem started a catering service. Bheem managed the business for 4 months more than the time for which Nala did and Nala invested Rs. 20,000 more than what Bheem did. At any given time only one of them manages the business though the service remains operational for the whole year. Nala and Bheem share the annual profit of Rs. 24,000 equally. If the person who managed the business is paid Rs. 1000 as salary per month. Then find the capital invested by Bheem.
 (1) Rs. 60,000 (2) Rs. 40,000
 (3) Rs. 20,000 (4) Can't determine
51. Find the single discount equivalent to a series discount of 20%, 10% and 5%.
 (1) 31.6% (2) 35%
 (3) 30% (4) 32%
52. The ratio of the cost of a clock to that of a wristwatch is 0.2. If the cost of the wristwatch increases to $\frac{5}{4}$ th of its value and the ratio of the costs increases by 10 percentage points, find the percentage increase in the cost price of the clock.
 (1) 75% (2) 85%
 (3) 87.5% (4) 82.5%
53. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is -
 (1) No profit, no loss (2) 5%
 (3) 8% (4) 10%
54. 30% of a number when subtracted from twice the number equals to 33 less than five times the same number. What is the number?
 (1) 15 (2) 10
 (3) 20 (4) 5
55. After getting two successive discounts, a shirt with a list price of Rs. 150 is available at Rs. 105. If the second discount is 12.5%, find the first discount.
 (1) 20% (2) 25%
 (3) 12% (4) 8%

56. If a dealer gets a commission of 10% on the list price from the company, then the profit made by the company is 50% of its manufacturing cost. If the dealer's commission is increased to 25%, then what will be the profit percentage on the cost of manufacturing for the company?
- (1) 10% (2) 25%
(3) 30% (4) 15%
57. An uneducated retailer marks all his goods at 50% above the cost price and thinking that he will still make 25% profit, offers a discount of 25% on the marked price. What is his actual profit on the sales?
- (1) 12.5% (2) 10%
(3) 15% (4) 20%
58. In the second and the third match of a series David's score increases by 10% and $21\frac{7}{33}\%$ respectively over the previous match. By what percentage did his score increase in the third match as compared to the first match?
- (1) $33\frac{1}{3}\%$ (2) $31\frac{7}{33}\%$
(3) $35\frac{15}{33}\%$ (4) $37\frac{9}{33}\%$
59. A dishonest dealer professes to sell his goods at cost price but uses a weight of 960 gms for a kg weight. Find his gain percent.
- (1) $4\frac{1}{6}\%$ (2) 5%
(3) $3\frac{1}{3}\%$ (4) $4\frac{1}{2}\%$
60. A milk vendor purchases milk at the rate of Rs. 8 per litre. He mixes one-fourth litre of water for every litre of milk. Find his profit percentage if he sells the diluted milk at Rs. 10 per litre.
- (1) 50% (2) 75%
(3) $66\frac{2}{3}\%$ (4) $56\frac{1}{4}\%$
61. A retailer buys 40 pens at the marked price of 36 pens from a wholesaler. If he sells these pens giving a discount of 1%, what is the profit percent?
- (1) 9% (2) 10%
(3) $11\frac{1}{9}\%$ (4) 12%
62. A man borrows Rs 6000 at 5% interest, on reducing balance, at the start of the year. If he repays Rs 1200 at the end of each year, find the amount of loan outstanding, in Rs. at the beginning of the third year.
- (1) 3162.75 (2) 4125.00
(3) 4155.00 (4) 5100.00
63. A shopkeeper sells some articles at the profit of 25% on the original price. What is the exact amount of profit? To find the answer, which of the following information given in Statements I and II is/are necessary?
I. Sale price of the article II. Number of articles sold
- (1) Only I is necessary
(2) Only II is necessary
(3) Either I or II is necessary
(4) Both I and II are necessary
64. A cellphone is available for Rs. 39000 cash or Rs. 6000 cash down payment followed by 4 equal monthly installments at 24% p.(1) compound interest, compounded monthly. Find the value of each installment.
- (1) 12000 (2) 15320
(3) 9640 (4) None
65. At what percentage above the CP must an article be marked so as to gain 33% after allowing a customer a discount of 5%?
- (1) 38% (2) 40%
(3) 45% (4) 50%
66. Rehman buys a few apples at 15 for a rupee and the same number of apples at 20 for a rupee. He mixes the two lots and sells them at 35 for 2 rupees. What is his gain or loss percentage?
- (1) 3.62% loss (2) 2.04% profit
(3) No profit, no loss (4) 2.04% loss
67. If the manufacturer gains 10%, the wholesale dealer 15% and the retailer 25%, then find the cost of production of a table, the retail price of which is Rs. 1265?
- (1) Rs. 800 (2) Rs. 850
(3) Rs. 900 (4) Rs. 950
68. A trader marked the price of his goods 20% more than that of the cost price. He then sells $\frac{1}{4}$ of his stock at a discount of 10%, and half of the stock at the marked price, and the rest at a discount of 25%. Find his gain percentage.
- (1) 18% (2) 9.5%
(3) 12% (4) 2.25%
69. When a producer allows 36% commission on the retail price of his product, he earns a profit of 8.8%. What would be his profit percent if the commission is reduced by 24%?
- (1) 23.48% (2) 26.5%
(3) 28% (4) 30%
70. The aggregate cost of 2 apples, 3 mangoes and 4 oranges is Rs. 6 more than the aggregate cost 1 apple, 2 mangoes and 2 oranges, and the aggregate cost of 1 apple, 2 mangoes and 1 orange is Rs. 8 less than the aggregate cost of 3 apples, 3 mangoes and 5 oranges. If the aggregate list price of 3 apples, 1 mango and 6 oranges is Rs. 12, then find how much percent discount one should give on the sale of 3 apples, 1 mango and 6 oranges such that there is a net profit of 5%?
- (1) 10% (2) 7.5%
(3) 15% (4) 12.5%
71. A shopkeeper sells some toys at Rs. 250 each. What percent profit does he make? To find the answer, which of the following information given in Statements I and II is/are necessary?
- I. Number of toys sold.
II. Cost price of each toy.
- (1) Only I is necessary
(2) Only II is necessary
(3) Both I and II are necessary
(4) Either I or II is necessary

72. Mr Alphonso buys 2 lots of mangoes. The first lot costs him Rs. 30 per dozen and the second lot Rs. 50 per dozen. He sold all the mangoes at Rs. 40 per dozen. Find his profit or loss percentage if he spent the same amount of money on each of the lots.
 (1) 6.66% loss (2) 6.67% profit
 (3) 5% profit (4) 5% loss
73. Monika purchased a pressure cooker at $\frac{9}{10}$ th of its selling price and sold it at 8% more than its SP find her gain percent.
 (1) 18% (2) 20%
 (3) 22% (4) 25%
74. A retail bookseller buys books at 58% of the list price. He expects to earn a minimum of 20% net profit on his selling price. What is the maximum discount that he can offer to his customer?
 (1) 31.6% (2) 25.5%
 (3) 27.5% (4) 30%
75. A man mixes two types of rice (X and Y) and sells the mixture at the rate of Rs. 17 per kg. Find his profit percentage.
 I. The rate of X is Rs. 20 per kg.
 II. The rate of Y is Rs. 13 per kg.
 (1) I alone sufficient while II alone not sufficient to answer
 (2) II alone sufficient while I alone not sufficient to answer
 (3) Either I or II alone sufficient to answer
 (4) Both I and II are not sufficient to answer
76. If a shopkeeper offers a discount of 20% on the list price of a washing machine, then he makes a profit of 12%. What is the percentage profit or loss, if he sells at a discount of 25% on the list price?
 (1) 0.6% loss (2) 0.5% profit
 (3) 4.25% loss (4) 5% profit
77. An article is sold at a certain price. By selling it at $\frac{2}{3}$ of that price one loses 10%. Find the gain percent at original price.
 (1) 30% (2) 33.33%
 (3) 35% (4) 40%
78. By selling 6 dozen oranges, a person incurs a loss which is equal to selling price of 1 dozen oranges. Find his loss percentage?
 (1) 14.28% loss (2) 12.25% loss
 (3) 16.66% loss (4) 14.66% loss
79. By selling a product with 20% profit, how much profit was earned?
 I. The difference between cost and selling price is Rs. 40.
 II. The selling price is 120 percent of the cost price.
 (1) I alone sufficient while II alone not sufficient to answer
 (2) II alone sufficient while I alone not sufficient to answer
 (3) Either I or II alone sufficient to answer
 (4) Both I and II are not sufficient to answer
80. A television set listed at Rs. 3,200 is sold to a retailer at successive discounts of 25% and 15%. What is the final amount paid by retailer of that television set?
 (1) Rs. 2,700 (2) Rs. 2,040
 (3) Rs. 2,584 (4) None of these

Class Practice Exercise 17

Profit, Loss, Discount III

DIRECTIONS: For the following questions, four options are given. Choose the correct options.

1. A tradesman sold an article at a loss of 20%. If the selling price had been increased by Rs. 100, there would have been a gain of 5%. What was the cost price of the article?
 (1) Rs. 350 (2) Rs. 400
 (3) Rs. 450 (4) Rs. 500
2. Two horses were sold for Rs. 12,000 each, one at a loss of 20% and the other at a gain of 20%. The entire transaction resulted in
 (1) no loss no gain (2) loss of Rs. 1,000
 (3) gain of Rs. 1,000 (4) None of these
3. By selling an article what is the profit percent gained?
 I. 5% discount is given on list price.
 II. If discount is not given, 20% profit is gained
 III. The cost price of the articles is Rs. 5000. Which of the statement(s) is/are necessary to answer the question.
 (1) Only I and II (2) Only II and III
 (3) Only I and III (4) All I, II and III
4. A man sells an article at a profit of 25%. If he had bought it at 20% less and sold it for Rs. 10.50 less, he would have gained 30%. Find the cost price of the article.
 (1) Rs. 45 (2) Rs. 50
 (3) Rs. 55 (4) Rs. 60
5. What was the percentage of discount given?
 I. 23.5% profit was earned by selling an almirah for Rs. 12,350.
 II. If there were no discount, the earned profit would have been 30%.
 III. The cost price of the almirah was Rs. 10,000. Which of the statement(s) is/are necessary to answer the question.
 (1) Only I and II (2) Only II and III
 (3) Only I and III (4) Any two of the three
6. The price of a jewel, passing through three hands, rises on the whole by 65%. If the first and the second seller earned 20% and 25% profit respectively, find the percentage profit earned by the third seller.
 (1) 5% (2) 10%
 (3) 12% (4) 15%
7. What is the percent profit earned by the shopkeeper on selling the articles in his shop?
 I. Labeled price of the articles sold was 130% of the cost price.
 II. Cost price of each article was Rs. 550.
 III. A discount of 10% on labeled price was offered. Which of the statement(s) is/are necessary to answer the question?
 (1) Only I (2) Only II
 (3) I and III (4) All the three are required
8. A man sold two flats for Rs. 6,75,958 each. On one he gains 16% while on the other he loses 16%. How much does he gain or loses in whole transaction?
 (1) No gain, no loss
 (2) Loss of Rs. 35,518.31
 (3) Loss of Rs. 43,260.00
 (4) Gain of Rs. 35,518.31
9. A man buys an article for Rs. 27.50 and sells it for Rs. 28.60. Find his gain percent.
 (1) 4% (2) 3.85%
 (3) 25% (4) 1.1%
10. A dealer sold three-fourth of his articles at a gain of 20% and the remaining at cost price. Find the gain earned by him in the whole transaction.
 (1) 15% (2) 20%
 (3) 13.33% (4) 5%
11. If a radio is purchased for Rs. 490 and sold for Rs. 465.50, find loss percent.
 (1) 5% (2) 5.26%
 (3) 4.6% (4) 24.5%
12. A man bought a horse and a carriage for Rs. 3000. He sold the horse at a gain of 20% and the carriage at a loss of 10%, thereby gaining 2% on the whole. Find the cost of the horse.
 (1) Rs. 1200 (2) Rs. 1800
 (3) Rs. 1000 (4) Rs. 1500
13. A person incurs 5% loss by selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit.
 (1) Rs. 1260 (2) Rs. 1200
 (3) Rs. 1334 (4) Rs. 1255
14. Find the single discount equivalent to a series discount of 20%, 10% and 5%.
 (1) 31.6% (2) 35%
 (3) 28% (4) 32.8%
15. After getting two successive discounts, a shirt with a list price of Rs. 150 is available at Rs. 105. If the second discount is 12.5%, find the first discount.
 (1) 20% (2) 25%
 (3) 7.5% (4) 10%
16. An uneducated retailer marks all his goods at 50% above the cost price and thinking that he will still make 25% profit, offers a discount of 25% on the marked price. What is his actual profit on the sales?
 (1) 12.5% (2) 25%
 (3) 18.75% (4) 10%
17. A book was sold for Rs. 27.50 with a profit of 10%. If it were sold for Rs. 25.75, then what would have been the percentage of profit or loss?
 (1) 3% profit (2) 3% loss
 (3) 7% loss (4) 2.91% profit

18. A retailer buys 40 pens at the marked price of 36 pens from a wholesaler. If he sells these pens giving a discount of 1%, what is the profit percent?
 (1) 10% (2) 9%
 (3) 9.09% (4) 11.11%
19. If the cost price is 96% of the selling price, then what is the profit percent?
 (1) 4.66% (2) 4%
 (3) 96% (4) 25%
20. At what percentage above the CP must an article be marked so as to gain 33% after allowing a customer a discount of 5%?
 (1) 40% (2) 38%
 (3) 34.74% (4) 30%
21. The CP of 21 article is equal to SP of 18 articles. Find the gain or loss percent.
 (1) 16.66% gain (2) 16.66% loss
 (3) 14.28% gain (4) 16% gain
22. When a producer allows 36% commission on the retail price of his product, he earns a profit of 8.8%. What would be his profit percent if the commission is reduced by 24%?
 (1) 23.49% (2) 24%
 (3) 32.8% (4) 31.2%
23. If books bought at prices ranging from Rs. 200 to Rs. 350 are sold at prices ranging from Rs. 300 to Rs. 425, what is the greatest possible profit that might be made in selling eight books?
 (1) Rs. 400 (2) Rs. 600
 (3) Rs. 800 (4) Rs. 1800
24. A shopkeeper sold an article for Rs. 2090.42. Approximately, what will be the percentage profit if he sold that article for Rs. 2602.58?
 (1) 15% (2) 20%
 (3) 25% (4) 30%
25. A shopkeeper sold an article for Rs. 2090.42. Approximately, what will be the percentage profit if he sold that article for Rs. 2602.58?
 (1) 15% (2) 20%
 (3) 25% (4) 30%
26. Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800 his gain percent is
 (1) 4% (2) $5\frac{5}{11}\%$
 (3) 10% (4) 12%
27. A fruit vendor buys certain numbers of oranges at 20 per Rs.15 and twice this number of oranges at 30 per Rs.20. He sells all the oranges at 24 per Rs.18. At most what discount percentage can he offer so that he does not has a loss?
 (1) 8% (2) 7
 (3) 7.14% (4) None of these
28. A shopkeeper purchased 70 kg of potatoes for Rs. 420 and sold the whole lot at the rate of Rs. 6.50 per kg. What will be his gain percent?
 (1) $4\frac{4}{7}\%$ (2) $6\frac{1}{4}\%$
 (3) $8\frac{1}{3}\%$ (4) 20%
29. Sam purchased 20 dozen of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his profit percent?
 (1) 3.5 (2) 4.5
 (3) 5.6 (4) 6.5
30. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit and loss is
 (1) 14 $\frac{2}{7}\%$ gain (2) 15% gain
 (3) 14 $\frac{2}{7}\%$ loss (4) 15% loss
31. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
 (1) Rs. 1090 (2) Rs. 1160
 (3) Rs. 1190 (4) Rs. 1202
32. By selling 33 meters of cloth, one gains the selling price of 11 meters. Find the gain percent.
 (1) 50% (2) 33.33%
 (3) 11% (4) 16.66%
33. If CP is 240 and profit is 15% then find SP?
 (1) Rs. 276 (2) Rs. 204
 (3) Rs. 264 (4) Rs. 360
34. If SP is 535 and CP is 500 then find profit %?
 (1) 7% (2) 8%
 (3) 9% (4) 10%
35. A sells an article which costs him Rs. 400 to B at a profit of 20%. B then sells it to C, making a profit of 10% on the price he paid to A. How much does C pay to B?
 (1) Rs. 472 (2) Rs. 476
 (3) Rs. 528 (4) Rs. 532
36. If CP is 340 and loss is 5% then find SP?
 (1) Rs. 323 (2) Rs. 357
 (3) Rs. 320 (4) Rs. 345
37. Peter purchased a machine for Rs. 80,000 and spent Rs. 5000 on repair and Rs. 1000 on transportation and sold it with 25% profit. At what price did he sell the machine?
 (1) Rs. 1,05,100 (2) Rs. 1,06,250
 (3) Rs. 1,07,500 (4) Rs. 1,17,500
38. If SP is 735 and loss is 2% then find CP?
 (1) Rs. 750 (2) Rs. 749.70
 (3) Rs. 700 (4) Rs. 755.70

39. By selling an article for Rs. 100, a man gains Rs. 15. Then, his gain% is
- (1) 15% (2) $12\frac{2}{3}\%$
- (3) $17\frac{11}{17}\%$ (4) $17\frac{1}{4}\%$
40. When an article is sold at a certain price, there is a profit of 28%.
- (1) Rs. 640 (2) Rs. 628
- (3) Rs. 528 (4) Rs. 360
41. When a commodity is sold for Rs. 34.80, there is a loss of 2%. What is the cost price of the commodity?
- (1) Rs. 35.51 (2) Rs. 43
- (3) Rs. 43.20 (4) Rs. 46.40
42. When CP and SP both are increased by 42%, then which of the following is true
- (1) It remains the same as the original profit percentage.
- (2) It increases by 42% of the original profit percentage.
- (3) It is equal to 42%
- (4) It increases by 42 percentage points.
43. If markup is x% and discount is x%
- Quantity A:** CP
- Quantity B:** SP
- (1) If Quantity A is greater
- (2) If Quantity B is greater
- (3) If both Quantity A and B are equal
- (4) If both quantities can't be compared using available data
44. If markup is x% and discount is y% and profit is z% (x, y, z are positive)
- Quantity A:** x
- Quantity B:** y
- (1) If Quantity A is greater
- (2) If Quantity B is greater
- (3) If both Quantity A and B are equal
- (4) If both quantities can't be compared using available data
45. Salesman sold a pen for ₹750 at 16.66% loss. Then find SP if profit is 22.22%.
- (1) Rs. 1100 (2) Rs. 1000
- (3) Rs. 1083.33 (4) Rs. 1050
46. CP of a pen is ₹320. Find the increase in SP if profit % is changed from 8% to 13%.
- (1) Rs. 16 (2) Rs. 25.60
- (3) Rs. 41.60 (4) Rs. 14.40
47. A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?
- (1) Rs. 18.20 (2) Rs. 70
- (3) Rs. 72 (4) Rs. 88.25
48. If CP of 10 pens is equal to SP of 12 pens, find profit or loss %.
- (1) 16.66% loss (2) 20% loss
- (3) 20% gain (4) 16.66% gain
49. The sale price of an article including the sales tax is Rs. 616. The rate of sales tax is 10%. If the shopkeeper has made a profit of 12%, then the cost price of the article is
- (1) Rs. 500 (2) Rs. 515
- (3) Rs. 550 (4) Rs. 600
50. CP of 30 pens is equal to SP of x pens and loss is 9.09%. Find x.
- (1) 33 (2) 30
- (3) 27 (4) 36
51. Saransh purchased 120 reams of paper at Rs. 80 per ream. He spent Rs. 280 on transportation, paid octroi at the rate of 40 paise per ream and paid Rs. 72 to the coolie. If he wants to have a gain of 8%, what must be the selling price per ream?
- (1) Rs. 86 (2) Rs. 87.48
- (3) Rs. 89 (4) Rs. 90
52. A salesman sold 44 articles and his profit is equal to the SP of 8 articles. Find profit %.
- (1) $22\frac{2}{9}\%$ (2) $18\frac{2}{11}\%$
- (3) 20% (4) 15%
53. A person bought 20 liters of milk at the rate of Rs. 8 per litre. He got it churned after spending Rs. 10 and 5 kg of cream and 20 litres of toned milk were obtained. If he sold the cream at Rs. 30 per kg and toned milk at Rs. 4 per litre, his profit in the transaction is
- (1) 25% (2) 35.3%
- (3) 37.5% (4) 42.5%
54. A man purchased 2 pens for ₹70 and ₹150 respectively. He sold the first pen at 40% profit and the second at 4% loss. Find his overall profit or loss %.
- (1) 10% profit (2) 10% loss
- (3) 12% profit (4) 15% profit
55. Jacob bought a scooter for a certain sum of money. He spent 10% of the cost on repairs and sold the scooter for a profit of Rs. 1100. How much did he spend on repairs if he made a profit of 20%?
- (1) Rs. 400 (2) Rs. 440
- (3) Rs. 500 (4) Rs. 550
56. If profit is 14.28%, find margin %.
- (1) 12.5% (2) 14.28%
- (3) 16.67% (4) 17.65%
57. A manufacturer undertakes to supply 2000 pieces of a particular component at Rs. 25 per piece. According to his estimates, even if 5% fail to pass the quality tests, then he will make a profit of 25%. However, as it turned out 50% of the components were rejected. What is the loss to the manufacturer?
- (1) Rs. 12,000 (2) Rs. 13,000
- (3) Rs. 14,000 (4) Rs. 15,000

58. If margin is 60%, find profit %.
 (1) 150% (2) 60%
 (3) 40% (4) 66.67%
59. A trader buys a chair for Rs. 600 and sell it for Rs. 765 at a credit of 4 months. Reckoning money worth 6% p.a. his gain percent is
 (1) 20% (2) 22.5%
 (3) 25% (4) 27.5%
60. If CP is 110 and mark-up is 36.36%, find MP.
 (1) Rs. 150 (2) Rs. 140
 (3) Rs. 139.96 (4) Rs. 145
61. A person sold his watch for Rs.24 and got a profit percentage equal to the cost price, then the cost price is
 (1) Rs.20 (2) Rs.18
 (3) Rs.22 (4) Rs.16
62. A shopkeeper increased the CP of a pen and marked the price as ₹160, then allowed a discount of 20%. Find SP of the pen.
 (1) Rs. 128 (2) Rs. 140
 (3) Rs. 136 (4) Rs. 160
63. An article was sold for Rs.y after giving a discount of x%. Then, its list price is
 (1) Rs. xy (2) $Rs. \frac{xy}{100}$
 (3) $Rs. \frac{100y}{100-x}$ (4) $Rs. \frac{(100-x)y}{x}$
64. If MP is ₹270 and SP is ₹240, find discount %.
 (1) 11.11% (2) 12.5%
 (3) 10% (4) 12%
65. By selling an article for Rs.816, a shopkeeper incurs a loss of 20%. At what price should he sell the article so as to gain 10%?
 (1) Rs.1,020 (2) Rs.1,122
 (3) Rs.1,120 (4) Rs.1,100
66. If mark-up is 200% and discount is 40%, find profit %.
 (1) 80% (2) 160%
 (3) 120% (4) 100%
67. A shopkeeper bought 60 pens at the rate of Rs.25 per pen. If the marked price of pens is Rs.30 per pen and the shopkeeper gives a discount of 12% on the marked price, then what will be the percentage profit gained by the shopkeeper?
 (1) 25% (2) 56%
 (3) 5.6% (4) 20%
68. If mark-up is 70% and profit is 19%, find discount %.
 (1) 30% (2) 51%
 (3) 19% (4) 50%
69. Suvakar was very much excited about an offer from Reebok that read "Buy things worth Rs.2,000 and get things worth Rs.1,000 absolutely free!". How much discount was there in the offer?
 (1) 50% (2) 100%
 (3) $33\frac{1}{3}\%$ (4) 25%
70. If mark-up is 40%, find discount % if profit is 0%.
 (1) $28\frac{4}{9}\%$ (2) 40%
 (3) 30% (4) 25%
71. By selling 6 dozen oranges a person loses selling price of 1 dozen oranges. Find his percentage loss.
 (1) 14.28% (2) $16\frac{2}{3}\%$
 (3) 10% (4) 20%
72. Two successive discounts of 20% and 35% are equivalent to a single discount of what %?
 (1) 48% (2) 55%
 (3) 42.5% (4) 60%
73. A dealer sells a set of furniture at 20% loss. Had he sold it for Rs.900 more, he would have made a 25% profit. At what price should he sell in order to make 20% profit?
 (1) Rs.1,800 (2) Rs.2,000
 (3) Rs.2,400 (4) Rs.2,800
74. A salesman gives a discount of 30% and still manages a profit of 33%. Find his mark-up %.
 (1) 90% (2) 63%
 (3) 43% (4) 70%
75. The cost price of a commodity is Rs.1,331. By selling it at a discount of Rs.100, the dealer makes a profit of 9.09%. If he decides not to give any discount, then what will be the profit made by him?
 (1) Rs.131 (2) Rs.121
 (3) Rs.221 (4) Rs.231
76. A milkman mixes 30% water in pure milk and sells at CP. Find his profit % and also find profit % if he sells at 10% less than CP.
 (1) 30% and 17% (2) 30% and 20%
 (3) 30% and 27% (4) 23.07% and 17%
77. The par value of the shares of company X and Y is Rs.10. The market price of the shares are Rs.40 and Rs.50 respectively. Find the ratio of the return on investment for an investor if the dividends are 20% and 40% respectively. Investment in both the cases is same.
 (1) 5 : 8 (2) 8 : 5
 (3) 8 : 13 (4) 13 : 8
78. Betal sold an article to Birbal at 20% profit, Birbal sold the same article to Chatur at 30% profit, and Chatur sold the same article to Tenali at 20% loss. If CP of Tenali is ₹1040, find the CP of Birbal.
 (1) Rs. 1000 (2) Rs. 1250
 (3) Rs. 1120 (4) Rs. 980
79. A cloth store is offering "Buy 3, get 1 free." What is the net percentage discount being offered by the store?
 (1) 25% (2) $33\frac{1}{3}\%$
 (3) 20% (4) 75%

80. A salesman bought every 5 articles for ₹4 and sold them every 4 articles for ₹5. Find his profit %.
- (1) 25% (2) 56.25%
(3) 36% (4) 64%
81. A man sells two items such that the cost price of one is equal to the selling price of the second, and the selling price of the first is equal to the cost price of the second. Which of the following is true?
- (1) He makes a profit
(2) He makes a loss
(3) He makes neither profit nor loss
(4) None of these
82. A salesman sold 2 articles — on the first there is a profit of 40% and on the second a loss of 40%. SP of both articles is same. Find overall loss %.
- (1) 16% loss (2) No profit no loss
(3) 20% loss (4) 20% profit
83. A book seller allows a discount of 10% on the marked price of a certain category of books which costs him Rs.600 per book. What price must be marked on the book to make a profit of 20%?
- (1) Rs.792 (2) Rs.800
(3) Rs.720 (4) Rs.726
84. A showroom offers “Buy 3 Get 2 Free” and the marked price is the same for all. Find the discount %.
- (1) 40% (2) 33.33%
(3) 50% (4) 66.67%
85. When an article is sold for Rs. X, loss percentage is equal to L%. However, when the same article is sold for Rs. Y, profit percentage is equal to P%. What is the cost price of that article?
- (1) Rs. $\frac{X-Y}{L+P}$ (2) Rs. $\frac{L+P}{X-Y}$
(3) Rs.100 (4) Rs.100
86. Ashok made a loss of 15% by selling 96 apples for ₹2040. How many apples must he sell for ₹2600 to make a 30% gain?
- (1) 80 (2) 100
(3) 65 (4) 104

Class Practice Exercise 18

Simple Interest, Compound Interest

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. What equal annual installment (in Rs.) will discharge a loan of Rs. 72800 borrowed at 20% pa. Compounded annually in 3 years?
 (1) 34,560 (2) 35,560
 (3) 35,580 (4) 34,480
2. A certain amount of money lent at compound interest becomes 2.25 times itself in 3 years. If the rate of interests for the first, second and the third years are in the ratio 4 : 5 : 10, find the rate of interest for the third year given that the interest is compounded annually?
 (1) 30% (2) 45%
 (3) 50% (4) 55%
3. A sum of Rs. P is to be repaid in two equal annual installments of Rs. 14520 each at 10% pa with interest being compounded annually. Another sum of Rs. Q is to be repaid at the same rate of simple interest in two equal annual installments of Rs. 14520. What is the difference between P and Q?
 (1) Rs. 100 (2) Rs. 150
 (3) Rs. 210 (4) Rs. 180
4. A sum becomes Rs. 20385 in 16 years at 6.25% p.a. with interest being compounded every moment. Find the sum (in Rs.). (Given $e = 2.718$)
 (1) 7250 (2) 7500
 (3) 7750 (4) 7800
5. Ajay deposited at certain sum in a fixed deposit with interest offered being compounded annually. The interests on the sum for the 3rd and 4th year differ by Rs. 145.20. The sum of the interests for the third year and the fourth year is Rs. 3049.20. Find the interest accrued (in Rs.) on it in the first 3 years.
 (1) 3672 (2) 3972
 (3) 3872 (4) 4072
6. The difference between the compound interest for the first year and second year on a certain sum at a certain rate of interest is Rs. 480. If the rate of interest is doubled, what will the difference (in Rs.) between the compound interest for the first year and the second year be?
 (1) 960 (2) 1440
 (3) 480 (4) 1920
7. Raghu borrowed Rs. 4000 at simple interest payable at the end of 7 years. After 2 years. He repaid Rs. 3000 towards the principal and after 5 more years, he repaid Rs. 3600 to settle his loan. Find the rate of interest at which he borrowed the sum.
 (1) 10% pa (2) 12.5% pa
 (3) 20% pa (4) 16.66% pa
8. Find the ratio of the 4 pans into which a certain sum is to be divided such that when the first, second, third and fourth parts are invested for 2 years, 3 years. 4 years and 5 years respectively at the same rate of simple interest, equal interests are obtained.
 (1) 30:20:15:12 (2) 12:15:20:30
 (3) 2:3:4:5 (4) 5:4:3:2
9. Ramu borrowed Rs. 62500 from a bank at 20% pa with interest being compounded annually. He repaid it in two annual installments. The amount he paid at the end of the second year was Rs. 2000 more than that paid at the end of the first year. Find the amount (in Rs.) he paid at the end of the first year.
 (1) 37500 (2) 42000
 (3) 35000 (4) 40000
10. The simple interest on a sum of money for 2 years at a certain rate of interest is Rs. 320. The interest on the same sum for the same duration and at the same rate of interest is Rs. 384, with interest being compounded annually. Find the sum (in Rs.).
 (1) 200 (2) 250
 (3) 300 (4) 400
11. A sum of money compounded annually becomes Rs. 625 in 2 years and Rs. 675 in 3 years. The rate of interest per annum is
 (1) 7% (2) 10%
 (3) 5% (4) 8%
12. Difference between compound interest and simple interest for 2 years on a sum of money is Rs. 160. If total simple interest is Rs. 2,880, then find rate of interest.
 (1) $5\frac{5}{9}\%$ (2) $12\frac{1}{2}\%$
 (3) $11\frac{1}{9}\%$ (4) 9%
13. After how many years (approximately) would the amount payable on a loan be twice the principal, if principal is lent at 20% CI, compounded half yearly?
 (1) 8 years (2) 6 years
 (3) 4 years (4) None of these
14. If the difference between the CI and the SI at the end of 2 years is Rs. 100, what is the principal? Rate is 5% per annum in both the cases. (Assume same principal for both the cases.)
 (1) Rs. 50,000 (2) Rs. 40,000
 (3) Rs. 10,000 (4) None of these
15. The compound interest on a certain sum of money for 2 years at 10% per annum is Rs. 420. The simple interest on the same sum at the same rate and for the same time will be
 (1) Rs. 350 (2) Rs. 375
 (3) Rs. 400 (4) None of these
16. In a certain sum simple interest for 4 years is 144 then simple interest on same sum for 9 years is
 (1) 324 (2) 288
 (3) 576 (4) 432

17. A sum of 450 becomes 750 in 2 years at $r\%$ rate of interest under simple interest then find r ?
- (1) 33.33% p.a. (2) 16.66% p.a.
(3) 66.66% p.a. (4) 40% p.a.
18. A certain sum becomes $\frac{17}{9}$ times in 4 years under simple interest. Find rate of interest?
- (1) 9.09% (2) 22.22 %
(3) 11.11% (4) None of these
19. A borrowed ₹10000 in the month of January which is absolutely interest free. He decided to pay back $x\%$ of the borrowed sum every year then find x if it took 25 years to clear the loan?
- (1) 5 (2) 4.5
(3) 4 (4) 5.5
20. If amount is 154% more than principal in 14 years under simple interest. Find rate of interest?
- (1) 11% pa (2) 15.4% pa
(3) 25.4% pa (4) 14% pa
21. A certain sum becomes 9100 in 3 years and 11900 in 7 years under simple interest then find principal and rate of interest?
- (1) 7000, 10% (2) 7000, 7%
(3) 7700, 10% (4) 7000, 8%
22. How long will it take for a sum of money to grow from Rs.1250 to Rs.10,000, if it is invested at 12.5% p.a. at simple interest?
- (1) 56 (2) 64
(3) 80 (4) 48
23. A sum of money at CI grows to 3 times in 7 years. In how many years will it grow to 27 times?
- (1) 21 (2) 17
(3) 19 (4) 28
24. A certain sum becomes 2.197 times in 3 years under compound interest. Find rate of interest compounded annually?
- (1) 30% (2) 13%
(3) 24% (4) None of these
25. Find compound interest on 2400 for 2 years if rate of interest are 10% p.a. and 15% p.a. for first and second years respectively?
- (1) Rs. 636 (2) Rs. 600
(3) Rs. 624 (4) Rs. 630
26. A sum ₹P becomes 5000 when it is invested for 5 years at $x\%$ p.a. under simple interest and sum ₹P becomes 5000 when it is invested for 5 years at $y\%$ p.a. under compound interest?
- Quantity A:** x
Quantity B: y
- (1) If Quantity A is greater
(2) If Quantity B is greater
(3) If Quantity A and B are equal
(4) If A and B are not comparable
27. What is the ratio of the compound interest and SI received after 2 years on a certain sum at 40% p.a.?
- (1) 7 : 5 (2) 6 : 5
(3) 5 : 4 (4) 5 : 6
28. If the ratio of the compound interest earned in n th and $(n+1)$ th year is 9:11. Find rate of interest?
- (1) $22\frac{2}{9}\%$ (2) 20%
(3) 18% (4) 25%
29. Find compound interest on 1200 at 20% p.a. compounded half yearly for 1 year?
- (1) Rs. 252 (2) Rs. 240
(3) Rs. 264 (4) Rs. 246
30. Difference between compound interest and simple interest for first 2 years is Rs64 and simple interest for 2 years is 640. Find principal and rate of interest if both are same for SI as well as CI?
- (1) Rs. 1500, 20% (2) Rs. 1600, 25%
(3) Rs. 1600, 10% (4) Rs. 1600, 20%
31. What rate of interest of simple interest would yield the same amount as that got at compound interest of 10% p.a. when the same principal is kept for 2 years in both the cases?
- (1) 10% pa (2) 10.5% pa
(3) 11% pa (4) 20% pa
32. The Amount invested by Dan in Bank of America is \$2000. The bank provides 10% rate growing annually. The Bank forms the following equation to determine the total amount that could be received by Dan at the end n th year:
 $P = 2000 (r)^n$ What is the value of r ?
- (1) 10 (2) 1.10
(3) 0.10 (4) 0.90
33. If a certain investment grows an annual interest of 8 percent compounded quarterly. The initial deposit was \$10 and the expression $\$10(x)^{4T}$ is used to find the value in account after T years. What is the value of x in the expression?
- (1) 1.08 (2) 1.2
(3) 1.02 (4) 1.8
34. A certain sum becomes 5 times in 20 years then in how many years same sum will become 10 times under SI
- (1) 40 (2) 45
(3) 50 (4) 90

Class Practice Exercise 19

Work, Time and Rate

DIRECTIONS: For the following questions, four options are given.

1. The average of the heights of the students in a class will become 180 cm if three students of the average height of 160 cm join the class and five students of the average height of 168 cm leave the class. If two students whose heights are 172 cm and 182 cm join the class, what will happen, to the average height of the class?
 - (1) It will increase
 - (2) It will decrease
 - (3) It will not change
 - (4) Can't be determine
2. 3 pumps, working 8 hours a day, can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day?
 - (1) 9
 - (2) 10
 - (3) 11
 - (4) 12
3. A job which 94 humans take 87 days to complete is given to some robots. The first day one robot whose rate of doing work is same as that of a human starts the work. If, on each day, starting from the second day one more robot joins the team of robots, then find in how many days the job gets completed, given that the rate of work of i th robot ($i > 1$) is twice that of the $(i-1)$ th robot?
 - (1) 12 days
 - (2) 10 days
 - (3) 11 days
 - (4) 13 days
4. P, Q and R can complete a job working together in 18 days. If P and Q together can complete the work in 24 days. Q and R in 36 days, then in how many days can P, Q and R complete the job Individually?
 - (1) 18, 54, 54
 - (2) 36, 54, 72
 - (3) 36, 72, 54
 - (4) 36, 72, 72
5. Howard and John take 12 days and 16 days respectively to complete a job. Gerard is at least as fast as John but atmost as fast as Howard. Gerard and John work on alternate days and completed the job in x days. Which of the following can be the value of x ?
 - (1) 12
 - (2) 13
 - (3) 14
 - (4) 17
6. A can copy 100 pages in 5 hr. A and B together can copy 100 pages in 4 hr. In what time can B copy 20 pages?
 - (1) 5 hr
 - (2) 4 hr
 - (3) 8 hr
 - (4) 2 hr
7. A takes twice as long as (B+C), and B takes thrice as long as (A+C). B completes a job in 72 days. If A increases his rate by 50% and C by 80%, how many days will A, B, and C together take to finish four times that job?
 - (1) 12 days
 - (2) 24 days
 - (3) 48 days
 - (4) 36 days
8. A can complete one-fifth of a job in 8 days. B can complete one-sixth of the job in 5 days. C can complete twice the job in 40 days. All the three work together to complete the job and they are paid a total of Rs. 260. Find the share of the fastest worker of the three
 - (1) Rs. 100
 - (2) Rs. 120
 - (3) Rs. 140
 - (4) Rs. 130
9. A worker has to complete a job of 175 units. On each day starting from the second. He does 75% of the part of the job he did on the previous day. Find the number of days in which the job is completed if he did 35 units of work on the 3rd day?
 - (1) 3
 - (2) 4
 - (3) 5
 - (4) 15
10. If the cost of x metres of wire is d rupees, then what is the cost of y metres of wire at the same rate?
 - (1) Rs. (xy/d)
 - (2) Rs. (xd)
 - (3) Rs. (yd)
 - (4) Rs. (yd/x)
11. Alloy I contains 30% zinc, 40% copper and the remaining is gold. Alloy II contains, 40% tin and some zinc and gold. On mixing these two alloys in a certain ratio, alloy III, which has 30% zinc and 25% tin is obtained. If equal weights of both Alloy I and Alloy III are mixed, then what percentage of the Alloy formed will be copper?
 - (1) 15%
 - (2) 40%
 - (3) 27.5%
 - (4) 32.5%
12. A, B and C can complete a work in 6, 12 and 18 days respectively. A and B started the work and C Joined them after one day. B left just 2 days before the completion of the whole work. In how many days was the work completed?
 - (1) 5
 - (2) 4
 - (3) 3
 - (4) 6
13. Irfan, Jay and Kornal can complete a job in 1 day, 20 days and 10 days respectively. They started the work together but Irfan was unwell on the first day, he could not work at his normal capacity and left after a day. The other two completed the job. Jay was paid Rs. 17000 out of the total of Rs. 60000 paid to them. At what percentage of his normal capacity, did Irfan work on the first day?
 - (1) 10%
 - (2) 20%
 - (3) 15%
 - (4) 25%
14. Mr. Smith can build a shed in x hr. After 2 hr, he stops his work. What part of the shed is left unbuilt?
 - (1) $\frac{x-3}{x}$
 - (2) $\frac{x-4}{x}$
 - (3) $1+\frac{2}{x}$
 - (4) $\frac{x-2}{x}$

15. Pipes P, Q and R are fitted to a tank. Each of Q and R can fill the tank in 6 hours. The capacity of P, which is an emptying pipe, is half of Q. If P is fitted at one fourth of the height of tank from the base and all the pipes are opened simultaneously, the empty tank would be filled in hours.
- (1) 4 (2) $3\frac{3}{4}$
(3) $3\frac{1}{2}$ (4) 3
16. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 minutes?
- (1) 648 (2) 1800
(3) 2700 (4) 10800
17. A can do a piece of work in 18 days, B in 36 days and C in 54 days. A starts the work and is joined by B after 1 day, C joins them after 4 more days. In how many more days will the work be completed?
- (1) 11 (2) 9
(3) 8 (4) 6
18. The weighted average of the scores of all the students of three sections X, Y and Z of a class is $33\frac{1}{3}\%$ more than the average of the section X. The weighted average of sections Y and Z is $65\frac{5}{11}\%$. If section X has a strength of 40 and an average score of 45, what is the combined strength of the section Y and Z?
- (1) 60 (2) 75
(3) 110 (4) 85
19. A fort had provision of food for 150 men for 45 days. After 10 days, 25 men left the fort. The number of days for which the remaining food will last, is
- (1) $146/5$ (2) $149/4$
(3) 42 (4) 54
20. A, B and C can complete a job together in 15 days whereas B and C can complete the same work together in 30 days. After A has worked for 5 days and stopped. B took over and worked for 15 days. C completed the remaining work in 30 days. In how many days can each complete the work alone?
- (1) 30; 45; 90 (2) 30; 90; 45
(3) 45; 30; 90 (4) 45; 90; 30
21. Five men and two boys working together can do four times as much work per hour as a man and a boy can complete together in an hour. Compare the work of a man with that of a boy.
- (1) 1 : 1 (2) 1 : 2
(3) 2 : 1 (4) 3 : 1
22. 39 persons can repair a road in 12 days, working 5 hours a day. In how many days will 30 persons, working 6 hours a day, complete the work?
- (1) 10 (2) 13
(3) 14 (4) 15
23. The number of matches played by a cricketer in the year 2003 is one-fourth of the total number of matches played by him up to the end of 2002. His average score (i.e. runs scored per match played) up to the end of the year 2003 is four-thirds of his average up to the end of the year 2002. What is the ratio of the runs scored in the, year 2003 to that of the total scored up to the end of the year 2002?
- (1) 2 : 3 (2) 3 : 2
(3) 3 : 4 (4) 15 : 16
24. P, Q and R can do a piece of work in 18 days, 36 days and 54 days respectively. If Q and R leave 1 day and 5 days respectively, before the completion of work, in how many days was the work been completed?
- (1) 10 (2) 11
(3) $10\frac{7}{11}$ (4) $10\frac{9}{11}$
25. A local grocer mixed three qualities of coffee T1, T2 and T3 which he had bought at Rs. 30 per kg, Rs. 45 per kg and Rs. 60 per kg in the ratio 3 : 2 : 5 to form a mixture T4. He mixes T1, T2 and T3 in the ratio 5 : 2 : 3 to form another mixture T5. If he mixes T4 and T5 in the ratio 1 : 5 and sells it at 10% profit, what will be the selling price (in Rs/kg) of the mixture?
- (1) 43.30 (2) 45.30
(3) 49.30 (4) 47.30
26. Two pipes can fill a tank in 18 min and 36 min respectively and a waste pipe can empty 2 gallons per minute. If all the three pipes work together, then the tank is filled in 15 min. The capacity of the tank is:
- (1) 120 gallons (2) 60 gallons
(3) 240 gallons (4) 180 gallons
27. Three members, whose average age is $53\frac{1}{3}$ years leave club A to join another one and at the same time a group of 5 persons with an average age of 48 years join the club A. The average age at the members of club A remains unchanged after the above two changes. What was the initial average age of the club?
- (1) 46 years (2) 55 years
(3) 40 years (4) Can't determine
28. A man completes $5/8$ of a job in 10 days. At this rate, how many more days will it takes him to finish the job?
- (1) 5 (2) 6
(3) 7 (4) 7.5
29. Ravi can finish a piece of work in 21 days and Prem can finish the same work in half the time taken by Ravi. If they work together then, what part of the same work can they finish in a day?
- (1) $1/7$ (2) $1/5$
(3) $1/4$ (4) $1/3$

30. P worked on a job for 4 hours and then Q joined him. After 8 more hours, P stopped working and Q took 34 more hours to complete the remaining part of the job. If P and Q together can complete the job in 24 hours. How much time will each of them individually take to complete the job? (both in hours)
- (1) 40, 60 (2) 48, 60
(3) 60, 45 (4) 40, 30
31. If a quarter kg of potato costs 60 paise, how many paise will 200 gm cost?
- (1) 48 paise (2) 54 paise
(3) 56 paise (4) 72 paise
32. There is a bottle full of pure listrene. One-third of the bottle is replaced with water. Find the ratio of water and listerine if the above operation is carried out for a total of four times.
- (1) 16 : 65 (2) 16 : 89
(3) 89 : 16 (4) 65 : 16
33. P, Q and R can complete a certain work in 8, 16 and 12 days respectively. If they work in rotation with one day each in the order of P, Q and R. In how many days will the work be completed?
- (1) $10\frac{2}{3}\%$ (2) $10\frac{3}{5}\%$
(3) 11 (4) 12
34. In a dairy farm, 40 cows eat 40 bags of husk in 40 days. In how many days one cow will eat one bag of husk?
- (1) 1 (2) $\frac{1}{40}$
(3) 40 (4) 80
35. Pipe A can fill a tank in 18 min; Pipe B can fill the tank in 27 min and Pipe C can empty the full tank in 36 min. If all of them work together, then find the time taken to fill the empty tank.
- (1) $16\frac{4}{7}$ minutes (2) 20 minutes
(3) 12 minutes (4) $15\frac{3}{7}$ minutes
36. Twelve hundred men can construct a wall of length 1500 metres in 30 days. In how many days can half the previous strength construct 1800 metres of wall working $\frac{1}{3}$ rd more time per day than in the previous case?
- (1) 54 (2) 48
(3) 36 (4) 27
37. A wheel that has 6 cogs is meshed with a larger wheel of 14 cogs. When the smaller wheel has made 21 revolutions, then the number of revolutions mad by the larger wheel is
- (1) 4 (2) 9
(3) 12 (4) 49
38. Two articles P and Q are sold at 20% loss and 10% profit respectively. resulting in an overall loss of 2.5%. If the cost prices are interchanged while the percentages of profit and loss on the two articles remain unchanged, then the overall percentage of profit or loss in the second case is how many times that of the first case? (Compare the magnitudes only)
- (1) 1 (2) 3
(3) 4 (4) 2
39. Fitty men can cut 600 trees in 12 days working 6 hrs/day. If 20 men leave the work after 6 days, how much time per day should the remaining men work to complete the job as per the original schedule?
- (1) 16 hours (2) 10 hours
(3) 6 hours (4) 4 hours
40. If 7 spiders make 7 webs in 7 days, then 1 spider will make 1 web in how many days?
- (1) 1 (2) $\frac{7}{2}$
(3) 7 (4) 49
41. A team of labourers promises to complete a piece of work in 10 days, but 5 of them do not turn up. If the rest of the team takes 12 days to complete the work, then find the original number of labourers.
- (1) 32 (2) 25
(3) 35 (4) 30
42. Six men and nine women can do a piece of work in 4 days. Four men and three women can do it in 8 days. In how many days can twenty men and six women do the same work?
- (1) 2 (2) 3
(3) 1 (4) 4
43. A flagstaff 17.5 m high casts a shadow of length 40.25 m. The height of the building, which casts a shadow of length 28.75 m under similar conditions will be -
- (1) 10 m (2) 12.5 m
(3) 17.5 m (4) 21.25 m
44. The average age of certain number of employees in an organization was 32 years. After one year if five new employees joined the organisation and the average age of the organization was still 32 years, then which of the following is definitely true?
- (1) The oldest employee in the office is in the new group.
(2) The aggregate age of new group is more than 160 years.
(3) The aggregate age of new group joined is less than 160 years.
(4) None of the above.
45. Three men or six women or nine boys can complete a job in 15 days. How many women should accompany 2 men and 3 boys to complete the same work in 5 days?
- (1) 10 (2) 12
(3) 9 (4) 15

46. In a camp, there is a meal for 120 men or 200 children. If 150 children have taken the meal, how many men will be catered to with remaining meal?
- (1) 20 (2) 30
(3) 40 (4) 50
47. Five men can complete a piece of work in 1 day whereas 4 women can complete the same piece of work in 3 days. How much time will a man and 2 women take to complete the same piece of work?
- (1) $1\frac{3}{7}$ days (2) $2\frac{8}{11}$ days
(3) $5\frac{5}{11}$ days (4) $2\frac{2}{9}$ days
48. One man and 3 boys take 3 times longer than 2 men and 12 boys to complete a job. How many boys should accompany 6 men such that a work is completed in half the time that 3 men and 12 boys would take?
- (1) 24 (3) 12
(3) 18 (4) None of these
49. An industrial loom weaves 0.128 metres of cloth every second. Approximately, how many seconds will it take for the loom to weave 25 metres of cloth?
- (1) 178 (2) 195
(3) 204 (4) 488
50. A group of 28 children have some money. Some children, who had a total of Rs. 1050 left the group. 7 other children who had an amount of Rs. 200 on average joined the group. As a result, the average amount with all the children did not change. If the children who left, rejoin the group the average amount with all the children would be Rs. 180 per head. Find the average amount with the original group of 28 children (in Rs./head)
- (1) 180 (2) 175
(3) 178 (4) 182
51. Three men and four women can together complete a work in 5 days, which 5 women and 6 boys can together complete in 4 days. How many boys are required to replace 30 men for completing a job?
- (1) 32 (2) 48
(3) 16 (4) 24
52. A and B can complete a job in 8 days working together. B alone can complete it in 12 days. Both of them worked together for 4 days and then B left. How long will A take to complete the remaining work?
- (1) $10\frac{3}{4}$ days (2) $11\frac{1}{2}$ days
(3) 16 days (4) 12 days
53. 36 men can complete a piece of work in 18 days. In how many days will 27 men complete the same work?
- (1) 12 (2) 18
(3) 22 (4) 24
54. Groundnuts contain 70% oil by weight. Oil is partially extracted and what is left behind is groundnut cake which contains 17% oil by weight. What is the weight of the groundnuts which yield 1 kilogram of cake?
- (1) 3.77 kg (2) 2.67 kg
(3) 3.58 kg (4) 2.77 kg
55. Wages for 30 women amount to Rs. 60,000 in 36 days. How many men must accompany 15 women to complete the work in 24 days given that a man earns double that of a woman? How much more amount is earned by men than women?
- (1) 10 men, Rs. 15,000
(2) 15 men, Rs. 20,000
(3) 15 men, Rs.30,000
(4) 10 men, Rs. 80,000
56. Ten men can complete a work in 18 days. After 6 days, 5 more men joined. In how many days will the remaining work be completed?
- (1) 6 days (2) 12 days
(3) 8 days (4) 10 days
57. 4 mat-weavers can weave 4 mats in 4 days. At the same rate, how many mats would be woven by 8 mat-weavers in 8 days?
- (1) 4 (2) 8
(3) 12 (4) 16
58. In two alloys iridium and vanadium are in the ratio 5 : 3 and 3 : 2 respectively. If 16 kg of first alloy and 15 kg of second alloy are mixed with a certain quantity of pure vanadium, then the resulting alloy consists of iridium and vanadium in equal proportions. The quantity of pure vanadium added is
- (1) 5 kg (2) 6 kg
(3) 7 kg (4) 9 kg
59. Two pipes can fill an empty cistern in 10 and 15 minutes respectively. A third pipe by itself can empty a full cistern in 5 minutes. In how much time will a cistern be emptied if the cistern was initially full and all three pipes are opened simultaneously?
- (1) 30 minutes (2) 60 minutes
(3) 45 minutes (4) 20 minutes
60. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is -
- (1) $\frac{1}{4}$ (2) $\frac{1}{10}$
(3) $\frac{7}{15}$ (4) $\frac{8}{15}$
61. A machine P can print one lakh books in 8 hours; machine Q can print the same number of books in 10 hours while machine R can print them in 12 hours. All the machines are started at 9 A.M. while machine P is closed at 11 A.M. and the remaining two machines complete work. Approximately at what time will the work (to print one lakh books) be finished?
- (1) 11:30 AM (2) 12 noon
(3) 12:30 PM (4) 1:00 PM

62. Nine years ago, the ages of the members of a family of seven people added up to 209 years. Three years later, one member died at the age of 52 years and a child was born. After two more years, one more member died again at 52 and another child was born. The current average age of all the members of the family is nearest to
- (1) 21 years (2) 24 years
(3) 23 years (4) 22 years
63. A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work?
- (1) 5 (2) 5.5
(3) 6 (4) 8
64. P, Q and R can do a piece of work in 18 days, 36 days and 54 days respectively. They start the work together but R and Q leave 5 days and 1 day respectively, before the completion of work. In how many days is the work completed?
- (1) 10 (2) 11
(3) $10\frac{7}{11}$ (4) $10\frac{9}{11}$
65. 4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?
- (1) 35 (2) 40
(3) 45 (4) 50

Class Practice Exercise 20

Time, Speed, Distance I

DIRECTIONS: For the following questions, four options are given. Choose the correct option.

1. The speeds of a scooter, a car and a train are in the ratio 1 : 4 : 16. If all of them cover equal distance, the ratio of time taken by each of the vehicle is:
 (1) 256 : 16 : 1 (2) 1 : 4 : 16
 (3) 16 : 4 : 1 (4) 16 : 1 : 4
2. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?
 (1) 3.6 (2) 7.2
 (3) 8.4 (4) 10
3. Excluding stoppages, the average speed of a bus is 54 km/hr, and including stoppages it is 45 km/hr. For how many minutes does the bus stop per hour?
 (1) 10 min (2) 9 min
 (3) 12 min (4) 7.5 min
4. By walking at $\frac{4}{5}$ th of his usual speed, a man reaches office 10 minutes later than usual. What is the usual time taken by the man to reach office?
 (1) 30 min (2) 40 min
 (3) 35 min (4) 50 min
5. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in $\frac{5}{3}$ hours, it must travel at a speed of -
 (1) 300 kmph (2) 360 kmph
 (3) 600 kmph (4) 720 kmph
6. Anu covers some distance at 4 km/hr and the rest at 6 km/hr. If the average speed for the entire journey is 5 km/hr and total distance covered is 50 km, then how much distance (in km) is covered at 4 km/hr?
 (1) 25 (2) 20
 (3) 30 (4) None of these
7. Walking at of his normal speed, Bharat takes 45 minutes more than the normal time to reach the office. Find his normal time.
 (1) $2\frac{1}{4}$ hr (2) 2 hr
 (3) 4 hr (4) 1.5 hr
8. If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is:
 (1) 50 km (2) 56 km
 (3) 70 km (4) 80 km
9. A person covers 180 km, 240 km and 300 km by car, train and bus respectively. Find the average speed of the whole journey if the speed of car, train and bus is 60 km/hr, 80 km/hr and 50 km/hr respectively.
 (1) $63\frac{1}{3}$ km/hr (2) 60 km/hr
 (3) $62\frac{1}{3}$ km/hr (4) 62 km/hr
10. A train travelling 100 km at x km/hr, arrives at its destination 1 hr late. What should have been the speed (in km/hr) for it to arrive on time?
 (1) $\frac{100x}{100+x}$ (2) $\frac{100x-x}{100x}$
 (3) $\frac{100x}{100-x}$ (4) None of these
11. A train can travel 50% faster than a car. Both start from point A at the same time and reach point B 75 kms away from A at the same time. On the way, however, the train lost about 12.5 minutes while stopping at the stations. The speed of the car is:
 (1) 100 kmph (2) 110 kmph
 (3) 120 kmph (4) 130 kmph
12. A car travels first 45 km of a journey at a speed of 15 km/hr. It covers the next 50 km at a speed of 25 km/hr and the rest 25 km at a speed of 15 km/hr. What is the average speed (in km/hr) of the car during the journey?
 (1) 18 kmph (2) 20 kmph
 (3) 18.33 kmph (4) 17.3 kmph
13. If a man walks from his house at 5 km/hr, he reaches office 5 minutes late. If he walks at 6 km/hr, he reaches 5 minutes early. Then the distance from his house to the office is,
 (1) 3 km (2) 4 km
 (3) 5 km (4) 6 km
14. Excluding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?
 (1) 9 (2) 10
 (3) 12 (4) 20
15. A car travels from A to B in 46 min. If the distance between A and B is 52.8 km, what is the (approximate) average speed of the car?
 (1) 1.14 km/hr (2) 40.5 km/hr
 (3) 68.86 km/hr (4) 84.2 km/hr
16. A train travels at a uniform speed of 15 km/hr. If the speed had been 10 km/hr more, it would have taken 8 hours less to cover the distance between stations A and B. What is the distance (in km) between A and B?
 (1) 100 (2) 300
 (3) 250 (4) 200
17. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is:
 (1) 1.5 hour (2) 2.5 hours
 (3) 3.5 hours (4) 4.5 hours

18. The shortest side 720 m remains fixed. The 1,350 m side is reduced by 60%. The area "cut off" from the triangle to accommodate the parking lot is itself a smaller triangle (or a portion of the original). Because the height 720 m remains the same and only the base is shortened, the area removed is the area of the triangle formed by the reduction in the base.
- (1) 0.39 km² (2) 0.26 km²
(3) 0.29 km² (4) 0.45 km²
19. If a man cycles at 10 km/hr, then he arrives at a certain place at 1 p.m. If he cycles at 15 km/hr from the same starting point, he will arrive at the same destination at 11 a.m. At what speed must he cycle to get there at noon?
- (1) 11 km/hr (2) 12 km/hr
(3) 13 km/hr (4) 14 km/hr
20. A man completes a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.
- (1) 220 km (2) 224 km
(3) 230 km (4) 234 km
21. X owns a Ferrari, McLaren, and Honda. McLaren is twice as fast as Honda; Ferrari is twice as fast as McLaren. McLaren takes 30 minutes to reach the office. X commutes 5 days/week. If he drove the Ferrari the first 2 days, in how many ways can he choose cars for the remaining 3 days so the total weekly commute is 5 hours?
- (1) 0 (2) 1
(3) 3 (4) 4
22. A Porsche has to cover 928 km in 10 hrs. If it covers first half of the journey in the of the total time, what should be its speed for the remaining journey so as it covers the entire journey in a specified time?
- (1) 124 km/hr (2) 164 km/hr
(3) 116 km/hr (4) 120 km/hr
23. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of the first train is:
- (1) 70 km/hr (2) 75 km/hr
(3) 84 km/hr (4) 87.5 km/hr
24. A 20km tunnel connects A and B with three gutters. G1 is 2.5km from A; G3 is 2.5km from B. Distance G1-G2 is half of G2-G3. An ambulance leaves A at 30km/hr, reaching G1 in 5 mins, then doubles its speed. If an accident occurs at G3 and the victim needs surgery within 40 mins, how much time does the doctor have? (Allow 1 min each for loading/unloading).
- (1) 4 min (2) 2.5 min
(3) 1.5 min (4) The patient died before reaching the hospital
25. A man leaves a point P at 6 a.m. and reaches the point Q at 10 a.m. Another man leaves the point Q at 8 a.m. and reaches the point P at 12 noon. At what time do they meet if they travel at a constant speed?
- (1) 9 a.m. (2) 10 a.m.
(3) 8.30 a.m. (4) 9 p.m.
26. A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is -
- (1) 35.55 km/hr (2) 36 km/hr
(3) 71.11 km/hr (4) 71 km/hr
27. Ram, Shyam and Hari went out for a 100 km journey. Ram and Hari started the journey in Ram's car at the rate of 25 kmph, while Shyam walked at 5 kmph. After sometime, Hari got off and started walking at the rate of 5 kmph and Ram went back to pick up Shyam. All three reached the destination simultaneously. The number of hours required for the trip was:
- (1) 8 (2) 6.4
(3) 7.5 (4) 4
28. A man travels three-fifths of the total distance from A to B at a speed 3a, and the remaining at a speed 2b. If he goes from B to A and returns at a constant speed 5c in the same time, then which of the following is true
- (1) $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$ (2) $a + b = c$
(3) $\frac{1}{a} + \frac{1}{b} = \frac{2}{c}$ (4) None of these
29. A car travelling with $\frac{5}{7}$ of its actual speed covers 42 km in 1 hr 40 min 48 sec. Find the actual speed of the car.
- (1) 125/17 km/hr (2) 25 km/hr
(3) 30 km/hr (4) 35 km/hr
30. The distance between A and B is 19 km. A cyclist starts from A at a constant speed towards (2) A car leaves from A 15 min later in the same direction. In 10 min it catches up with the cyclist and continues towards B; after reaching B, it turns around and in 50 min after leaving, car encounters the cyclist the second time. The speed of the car is
- (1) 30 km/hr (2) 40 km/hr
(3) 35 km/hr (4) 80 km/hr
31. A wheel rotates 10 times in a minute and moves 20 ft forward in each rotation. How much time does the wheel take if it moves 970 ft forward?
- (1) 2 min (2) 4.85 min
(3) 48.5 min (4) 9.7 min
32. In covering a distance of 30 km, Abhay takes 2 hours more than Sameer. If Abhay doubles his speed, then he would take 1 hour less than Sameer. Abhay's speed is:
- (1) 5 kmph (2) 6 kmph
(3) 6.25 kmph (4) 7.5 kmph
33. The distance between Ankit's house and office is 60 km. One day, he covers half of the distance at twice of his usual speed and the rest of the distance at half of his usual speed. If he took 75 minutes more than the usual time, find the usual speed (in km/hr).
- (1) 12 (2) 15
(3) 20 (4) 10

34. A train approaches a tunnel AB. Inside the tunnel is a cat located at a point that is $\frac{3}{8}$ of the distance AB measured from the entrance A. When the train whistles the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat at exactly the exit. What is the ratio of the speed of the train to that of the cat?
- (1) 3 : 1 (2) 4 : 1
(3) 5 : 1 (4) None of these
35. Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?
- (1) 8 kmph (2) 11 kmph
(3) 12 kmph (4) 14 kmph
36. A can cover a certain distance in 30 minutes and B can cover the same distance in 70 minutes. In how much time can C cover the same distance, if his speed is the sum of the speeds of A and B?
- (1) 35 minutes (2) 28 minutes
(3) 21 minutes (4) 42 minutes
37. A truck travelling at 70 km/hr uses 30% more diesel to travel a certain distance than it does when it travels at a speed of 50 km/hr. If the truck can travel 19.5 km/L of diesel at 50 km/hr, how far can the truck travel on 10 L of diesel at a speed of 70 km/hr?
- (1) 130 km (2) 140 km
(3) 150 km (4) 175 km
38. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the cars is:
- (1) 2 : 3 (2) 3 : 2
(3) 3 : 4 (4) 4 : 3
39. After travelling for 30 minutes, a train met with an accident and was stopped there for 45 minutes. Due to the accident, its speed reduced to of its initial speed and the train reached its destination 90 minutes late. Had the accident occurred 60 km after the point it occurred earlier, the train would have reached 30 minutes earlier than it was reaching after the previous accident. Find the length of the journey.
- (1) 90 km (2) 120 km
(3) 60 km (4) 180 km
40. The speed of a railway engine is 42 kmph when no compartment is attached, and the reduction in speed is directly proportional to the square root of the number of compartments attached. If the speed of the train carried by this engine is 24 kmph when 9 compartments are attached, then the maximum number of compartments that can be carried by the engine is
- (1) 49 (2) 48
(3) 46 (4) 47
41. A farmer travelled a distance of 61 km in 9 hours. He travelled partly on foot @ 4 km/hr and partly on bicycle @ 9 km/hr. The distance travelled on foot is -
- (1) 14 km (2) 15 km
(3) 16 km (4) 17 km
42. Train X departs from station A at 11 a.m. for station B, which is 180 km away from A. Train Y departs from station B at 11 a.m. for station A on the same day. Train X travels at an average speed of 70 km/hr and does not stop anywhere until it arrives at station B. Train Y travels at an average speed of 50 km/hr, but has to stop for 15 minutes at station C, which is 60 km away from station B enroute to station A. Ignoring the lengths of the trains, what is the distance, to the nearest integer kilometre, from station A to the point where the trains cross each other?
- (1) 112 km (2) 118 km
(3) 120 km (4) None of these
43. A man covered a certain distance at some speed. Had he moved 3 kmph faster, he would have taken 40 minutes less. If he had moved 2 kmph slower, he would have taken 40 minutes more. The distance (in km) is -
- (1) 35 (2) $110/3$
(3) 37.5 (4) 40